

Universal All-Weight Valuative Syntomic-to-Étale Comparison

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Abstract

Let V be a henselian valuation ring of mixed characteristic $(0, p)$. We prove, for every $n \geq 0$, that the mod- p syntomic-to-étale comparison

$$\mathbf{F}_p(n)_{\text{syn}}(V) \longrightarrow R\Gamma_{\text{ét}}(V[1/p], \mu_p^{\otimes n})$$

is an isomorphism in degrees $< n$ and injective in degree n . The proof reduces to complete rank-one valuation rings and has three new ingredients. First, a Frobenius–Witt calculation gives weight-uniform weak F -smoothness for every complete rank-one valuation ring. Second, a Tor-unital valuation double, together with degree-three Dennis–Stein operations and an all-degree normal-form argument, proves the Milnor endpoint in every weight. Third, after Frobenius base change, a geometric trivialization of the Breuil–Kisin line identifies all twisted Nygaard-completion defects; the endpoint and Bott localization then force those defects to vanish inductively. The argument includes $p = 2$, arbitrary value groups, and noncomplete valuation rings.

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1 Introduction

1.1 Main theorem

For a p -complete ring R write

$$S_n(R) = \mathbf{F}_p(n)_{\text{syn}}(R), \quad E_n(R) = R\Gamma_{\text{ét}}(R[1/p], \mu_p^{\otimes n}),$$

and let $\gamma_{R,n}: S_n(R) \rightarrow E_n(R)$ be the natural comparison. For a noncomplete valuation ring we use the standard extension of syntomic cohomology by the completion/generic-fiber pullback, as in [1].

Theorem 1.1 (Universal valuative comparison). *Let p be any prime and let V be a henselian valuation ring of mixed characteristic $(0, p)$. Then for every $n \geq 0$,*

$$\text{cofib}(S_n(V) \xrightarrow{\gamma_{V,n}} E_n(V)) \in \mathbf{D}^{\geq n}(\mathbf{F}_p).$$

Equivalently, $\gamma_{V,n}$ is an isomorphism on H^q for $q < n$ and injective on H^n .

The theorem is uniform in the prime, in the rank and value group of V , and in the weight. In particular no discreteness, no noetherian hypothesis, and no assumption $p > 2$ is made.

1.2 Proof architecture

The proof is organized so that each difficult point is isolated in a single reusable statement. We give the complete arguments for the new derived, valuative, and Milnor-theoretic steps; standard results from prismatic cohomology, syntomic cohomology, and valuation theory are cited in their precise form rather than reproved.

Step 1. Reduction and low-degree surjectivity. By the valuative excision theorem of Bachmann–Elmanto–Morrow, it suffices to treat complete rank-one valuation rings. For such rings, every generic Milnor symbol of cohomological degree $q < n$ is lifted through an explicit semistable regular local model. This gives surjectivity of $H^q S_n \rightarrow H^q E_n$ below the weight.

Step 2. Weak Frobenius smoothness. Let $A = V/p$. Combining the universal Nygaard–Sen triangle of Bhatt–Lurie with Mao’s Frobenius–Witt cotangent complex, we calculate the Frobenius pullback of the Bott defect

$$Q_i(V) = \operatorname{cofib}(\theta : \operatorname{gr}_N^i \Delta_V/p \rightarrow \operatorname{gr}_N^{i+p} \Delta_V/p).$$

For nondiscrete rank one it becomes a direct sum of flat exterior powers in degrees $i + 1, \dots, i + p$. Frobenius pullback reflects Tor lower bounds, hence V is weakly F -smooth in the sense of Bhatt–Mathew.

Step 3. Completion defects and first-overlap separation. Finite Bott strictness identifies the fiber of Bott multiplication on prismatic terms with a Hodge–Tate quotient. The induced twisted Nygaard filtration is shown to be separated on the first perfectoid Čech overlap. The key input is a uniform separation estimate for complete intersections with a fixed number of equations. Descent then gives

$$\mathcal{E}_k(V) := \operatorname{cofib}(\overline{\Delta}_V\{k\}/p \rightarrow \widehat{\Delta}_V\{k\}/p) \in \mathbf{D}^{\geq 1}$$

for every twist k .

Step 4. The Milnor endpoint. For a nondiscrete rank-one V with residue field k , the valuation double $T = V \times_k V$ is Tor-unital. Syntomic cohomology therefore satisfies excision on this square. A degree-three Dennis–Stein operation on the relative ideal is extended, using the double, to operations in all Milnor degrees. An abstract graph-normal-form lemma then proves

$$\widehat{K}_n^{\mathbf{M}}(V)/p \hookrightarrow K_n^{\mathbf{M}}(\operatorname{Frac} V)/p$$

for every n . This is exactly injectivity of the comparison in the endpoint degree.

Step 5. Twist periodicity after Frobenius base change. Put $\psi : V \rightarrow A$, $v \mapsto \bar{v}^p$. After derived pullback by ψ , the Breuil–Kisin line becomes trivial. The trivialization is obtained geometrically from the Hodge–Tate stack by mapping a two-dimensional regular prism

$$\mathbf{Z}_p[[T, B]]/(p - T^p B) \longrightarrow V.$$

Multiplication by the resulting unit identifies all $\psi^* \mathcal{E}_k$.

Step 6. Vanishing of all completion defects. Starting from $\mathcal{E}_k \in \mathbf{D}^{\geq 1}$, the truncated Bott-defect comparison, Bott localization, low-degree surjectivity, and the endpoint show that one twist has one additional vanishing degree. Frobenius-pulled twist periodicity propagates that vanishing to every twist. Induction gives $\mathcal{E}_k = 0$ for all k .

Step 7. Conclusion. Once the completion defects vanish, Bott multiplication is injective on syntomic cohomology through the weight. Bott localization identifies the telescope with generic étale cohomology, giving injectivity in all required degrees; Step 1 supplies surjectivity below the weight. The complete rank-one result is then transported back to arbitrary henselian valuation rings.

The logical dependence is therefore

weak F -smoothness	$\implies$	completed Bott control			
first-overlap separation	$\implies$	$\mathcal{E}_k \in D^{\geq 1}$			
valuation double	$\implies$	Milnor endpoint	$\implies$	$\mathcal{E}_k = 0$	$\implies$ Theorem 1.1.
BK-line unit	$\implies$	$\psi^* \mathcal{E}_k \simeq \psi^* \mathcal{E}_\ell$			

1.3 External inputs and conventions

We use prismatic and syntomic conventions from Bhatt–Lurie [2], the Hodge–Tate and Nygaard calculations of Bhatt–Scholze [7], and the Bott class and F -smoothness formalism of Bhatt–Mathew [4]. All tensor products, completions, fibers, and cofibers are derived unless flatness has been established. Complexes are cohomologically graded, so a module $M[-r]$ is concentrated in degree r .

For a p -quasisyntomic ring R , Δ_R denotes absolute prismatic cohomology, $N^{\geq r} \Delta_R$ its Nygaard filtration, and $\Delta_R\{k\}$ the k th Breuil–Kisin twist. A hat indicates Nygaard completion. We write $\overline{\Delta}_R\{k\} = \Delta_R\{k\}/I$ for the Hodge–Tate specialization, with the context determining the prism ideal I . The universal degree- p Nygaard class is denoted θ and the mod- p syntomic Bott class by v_1 ; its twist is $p - 1$.

2 Reduction to complete rank one and surjectivity below the weight

2.1 The valuative reduction

The following reduction is the form of [1, Remark 8.4] that we use; its proof rests on the v -excision square of [1, Proposition 7.25].

Proposition 2.1 (Bachmann–Elmanto–Morrow). *Fix p and n . If*

$$\mathrm{cofib}(S_n(W) \rightarrow E_n(W)) \in D^{\geq n}$$

for every p -complete henselian rank-one valuation ring W of mixed characteristic, then the same assertion holds for every henselian mixed-characteristic valuation ring.

Proof. We recall only the mechanism needed later. Completion does not change the comparison cofiber, by the completion/generic-fiber definition of syntomic cohomology. For a complete valuation ring V , let $\mathfrak{q} = \sqrt{pV}$. The localization $V_{\mathfrak{q}}$ has rank one and the same p -inverted fraction field. The v -excision square compares V , $V_{\mathfrak{q}}$, $V/\mathfrak{q}$ and $k(\mathfrak{q})$. The two characteristic- p terms have comparison cofibers in $D^{\geq n}$; in degree n this uses injectivity of logarithmic forms from a Cartier-smooth valuation ring to its fraction field. Hence the desired connectivity transfers from $V_{\mathfrak{q}}$ to V . See [1, Proposition 7.25 and Remark 8.4] for the complete argument. $\square$

From now until section 10 we work with a p -complete henselian rank-one valuation ring V of mixed characteristic and fraction field K .

2.2 A density lemma

Lemma 2.2. *Let $\Gamma \subset \mathbf{R}$ be a nondiscrete nonzero ordered subgroup. For every $m \geq 1$, every coset of $m\Gamma$ is dense in $\mathbf{R}$. Consequently, if K is a rank-one valued field with nondiscrete value group, every class in $K^\times/K^{\times m}$ has representatives of arbitrarily small positive valuation.*

Proof. A nondiscrete subgroup of $\mathbf{R}$ contains arbitrarily small positive elements, hence is dense. Multiplication by m and translation preserve density. If z represents the desired class and $\gamma \in \Gamma$ is chosen so that $v(z) + m\gamma$ lies in a prescribed interval, choose $h \in K^\times$ with $v(h) = \gamma$; then zh^m has the required value. $\square$

2.3 The perfectoid quasisyntomic cover

The prismatic arguments below use a single canonical cover of a complete rank-one valuation ring.

Lemma 2.3 (Perfectoid valuation cover). *Let C be the completion of an algebraic closure of K and put $W = \mathcal{O}_C$. Then $V \rightarrow W$ is a p -completely faithfully flat quasisyntomic cover. For every $a \geq 0$,*

$$A^a = (W^{\otimes_V(a+1)})_p^\wedge$$

is a discrete p -torsion-free quasiregular semiperfectoid ring. Absolute prismatic cohomology, its finite Nygaard quotients and Hodge–Tate specializations, and syntomic cohomology satisfy quasisyntomic descent on the resulting Čech nerve.

Proof. For an extension of valued fields, the cotangent complex of the extension of valuation rings has no homology above degree one and its degree-one homology is torsion-free; in characteristic zero the stronger perfect-base form applies. This is the valuation-theoretic cotangent theorem of Gabber–Ramero [8, Theorem 6.5.8]. Since torsion-free modules over a valuation domain are flat and every module over a valuation domain has flat dimension at most one, it gives the quasisyntomic Tor bound for $V \rightarrow W$. The valuation extension is faithfully flat, and p -adic completion preserves its reductions modulo p^r , so the completed map is a p -completely faithfully flat quasisyntomic cover.

The tensor powers are flat over V , hence p -torsion-free; their derived p -completions are discrete and remain p -torsion-free. Quasisyntomicity is preserved by the relevant base changes. To see the semiperfectoid condition, choose compatible p -power roots in the algebraically closed perfectoid valuation ring W of elements in the additional tensor factors. A perfectoid root-polynomial algebra over the first copy of W surjects modulo p onto the tensor product, and successive p -adic lifting gives a surjection onto its completion. Thus every A^a is QRSP.

Quasisyntomic descent for prismatic cohomology and the finite Hodge–Tate/Nygaard constructions follows from [7, Construction 7.6 and Theorem 12.2] and the absolute Nygaard descent results of [2, Section 5.5]. Syntomic descent then follows from its description as the Frobenius-minus-canonical-map fiber [2, Proposition 7.4.6]. $\square$

2.4 Prime-to- p cyclotomic descent

We will repeatedly adjoin μ_p . The point is that this can be undone by a finite-stage trace even when the integral closure is not finite over V .

Lemma 2.4. *Let V be henselian rank one, let L/K be cyclic of degree m prime to p , and suppose $\mu_m \subset K$. If V' is the valuation ring of L , then there is a finite étale valuation extension U/V such*

that V' is a filtered union of monogenic finite-free U -algebras, all having fraction field L . Their rank over V is m .

Proof. Choose a Kummer generator r with $r^m \in K$. Let e be the order of $v(r)$ in Γ_L/Γ_K and put $f = m/e$. After multiplying r^e by a scalar in $K^\times$, obtain a unit s satisfying $s^f \in V^\times$. Henselianity and the prime-to- p hypothesis show that $U = V[s]$ is finite etale of rank f , and a direct valuation-basis argument identifies it with the valuation ring of $K(s)$.

Now $L = \text{Frac}(U)(r)$ has degree e and the values of $1, r, \dots, r^{e-1}$ are distinct modulo Γ_K . In the nondiscrete case, after replacing r by br with $b \in K(s)^\times$ and $v(br) > 0$ arbitrarily small, any prescribed finite subset of V' lies in the monogenic order $U[br]$. These orders form a directed system. In the discrete case one chooses the Kummer generator of minimal positive value and obtains $V' = U[r]$. This proves the claim. $\square$

For a monic polynomial algebra $B = A[T]/(f)$ of degree d , syntomic Gysin maps and the projective bundle formula give a trace

$$\text{Tr}_f : S_i(B) \longrightarrow S_i(A)$$

whose composite with restriction is multiplication by d ; on a finite separable generic fiber it realizes etale corestriction. This follows from syntomic Gysin functoriality [16, Theorem 5.22] and the projective bundle formula [2, Theorem 9.3.1]. We will use this construction only for the monogenic orders of [theorem 2.4](#).

Proposition 2.5 (Cyclotomic descent). *Let $L = K(\mu_p)$ and let V' be its valuation ring. For every i, q :*

1. $H^q S_i(V) \rightarrow H^q S_i(V')$ is injective;
2. an etale class over K whose restriction to L has a syntomic lift over V' already has a syntomic lift over V ;
3. injectivity of $H^q S_i(V') \rightarrow H^q E_i(V')$ implies injectivity over V .

Proof. The degree $m = [L : K]$ divides $p - 1$. By [theorem 2.4](#), V' is a filtered union of finite-free monogenic stages B_t of total rank m over V , after a finite etale intermediate extension. Mod- p syntomic cohomology commutes with filtered colimits [2, Corollary 8.4.11]. Composing the two monogenic traces gives maps

$$T_t : H^q S_i(B_t) \rightarrow H^q S_i(V), \quad T_t \circ \text{res} = m,$$

compatible with generic corestriction. Since $m \in \mathbf{F}_p^\times$, (1) follows. If α_t lifts the restriction of an etale class β , then

$$\gamma(T_t \alpha_t) = \text{Cor}_{L/K} \text{Res}_{L/K}(\beta) = m\beta,$$

which proves (2) after multiplying by m^{-1} . Part (3) follows from (1). For $p = 2$ one has $m = 1$. $\square$

2.5 Semistable lifting below the weight

Proposition 2.6 (Surjectivity below the weight). *For every $i \geq 1$ and $0 \leq q < i$, the comparison*

$$H^q S_i(V) \longrightarrow H_{\text{et}}^q(K, \mu_p^{\otimes i})$$

is surjective.

Proof. By [theorem 2.5](#) we may assume $\mu_p \subset K$ and fix a primitive root ζ . If the valuation is discrete, V is a complete DVR and the assertion follows from the regular comparison theorem [[4](#), Theorem 4.15 and Proposition 5.2]. Assume the value group is nondiscrete and put $\lambda = \zeta - 1$.

By the norm-residue theorem [[18](#), Theorem 6.16], the target is generated by classes

$$\zeta^{\otimes(i-q)} \cup [a_1] \cup \cdots \cup [a_q].$$

Using [theorem 2.2](#), replace each a_r in its p th-power class so that

$$0 < v(a_r) < \frac{v(\lambda)}{q+1}.$$

Then $d = \lambda/(a_1 \cdots a_q)$ lies in the maximal ideal. Consider the henselian regular local ring

$$R = (\mathbf{Z}_p[\zeta][X_1, \dots, X_q, Y]/(X_1 \cdots X_q Y - \lambda))_{(X_1, \dots, X_q, Y)}^h.$$

It is regular of dimension $q+1$, and the assignments $X_r \mapsto a_r$, $Y \mapsto d$ define a local map $R \rightarrow V$. After inverting p the X_r are units. Every regular noetherian ring is F -smooth, and on a p -torsion-free F -smooth ring the syntomic-to-etale comparison is an isomorphism below the weight [[4](#), Theorem 4.15 and Proposition 5.2]. Since $q < i$, the displayed generic symbol has a syntomic lift on R , whose pullback gives the desired lift on V . The argument is unchanged for $p=2$, where $\lambda = -2$. $\square$

3 Weak Frobenius smoothness of complete rank-one valuation rings

The aim of this section is to prove a weight-uniform lower bound for the Bott defect on Nygaard graded pieces. The proof is completely different in the discrete and nondiscrete cases; the latter is the essential one.

3.1 The universal Bott-defect triangle

For an animated ring R , set

$$A_R = R \otimes_{\mathbf{Z}}^{\mathbf{L}} \mathbf{F}_p, \quad C_R = L_{A_R/\mathbf{F}_p}, \quad K_n(R) = L \bigwedge^n C_R[-n].$$

Write

$$G_n(R) = \mathrm{gr}_N^n \Delta_R/p, \quad Q_i(R) = \mathrm{cofib}(\theta : G_i(R) \rightarrow G_{i+p}(R)).$$

The Nygaard–Sen description of $\mathrm{gr}_N^n \Delta_R$ gives a fiber sequence

$$\mathrm{gr}_N^n \Delta_R \longrightarrow \mathrm{Fil}_n^{\mathrm{conj}} \Omega_R^{\mathrm{D}} \xrightarrow{\Theta+n} \mathrm{Fil}_{n-1}^{\mathrm{conj}} \Omega_R^{\mathrm{D}},$$

where the n th conjugate graded piece is $(L \bigwedge^n L_{R/\mathbf{Z}})_p^{\wedge}[-n]$ and Θ acts there by $-n$; see [[2](#), Construction 4.7.1 and Remark 5.5.8]. We also use the polynomial-variable decompositions of [[2](#), Example 5.6.10 and Variants 5.6.12–5.6.13].

Lemma 3.1 (Universal triangle). *For every animated ring R and $i \geq 0$ there is a natural fiber sequence in $D(A_R)$*

$$Q_i(R) \xrightarrow{q_i} K_{i+p}(R) \xrightarrow{\beta_i} K_i(R).$$

The construction is compatible with adjoining polynomial variables: under

$$K_n(R[X]) \simeq K_n(R)[X] \oplus K_{n-1}(R)[X][-1],$$

the map $\beta_i(R[X])$ decomposes as

$$\beta_i(R)[X] \oplus \beta_{i-1}(R)[X][-1].$$

Proof. The multiplicative Hodge projection gives maps $h_n : G_n(R) \rightarrow K_n(R)$. The Hodge image of θ is canonically zero, since it is pulled back from $\mathbf{Z}_p$ and $K_p(\mathbf{Z}_p) = 0$. Thus h_{i+p} descends to $q_i : Q_i \rightarrow K_{i+p}$.

We keep track of the map q_i and of its specified nullhomotopy before making any cohomological calculation. This point is needed later when the construction is animated and when polynomial variables are used to detect the scalar on exterior-power summands.

First suppose that R is a polynomial $\mathbf{Z}$ -algebra. Then the conjugate filtration is the Postnikov filtration. Writing $\widehat{\Omega}_R^r = (\Omega_{R/\mathbf{Z}}^r)_p^\wedge$, the Sen sequence gives, for $n > 0$,

$$H^s(\mathrm{gr}_N^n \Delta_R) = \begin{cases} \widehat{\Omega}_R^{s-1}/(n-s+1), & 1 \leq s < n, \\ \widehat{\Omega}_R^n, & s = n, \\ 0, & s \leq 0 \text{ or } s > n. \end{cases}$$

Indeed, in degree $s-1 < n$ the operator $\Theta + n$ is multiplication by the nonzero integer $n-s+1$ on the p -torsion-free module $\widehat{\Omega}_R^{s-1}$, and the top Hodge projection identifies the degree- n group.

Set $n = i + p$. After derived reduction modulo p , the only cohomology of G_{i+p} in degrees strictly larger than i occurs in degrees $i+1$ and $i+p$:

$$H^{i+1}(G_{i+p}) = \Omega_R^i/p, \quad H^{i+p}(G_{i+p}) = \Omega_R^{i+p}/p.$$

The first group occurs because among the integers $n-s+1$ with $i < s < n$, the unique multiple of p is p , at $s = i+1$; the additional Tor group produced by reducing that p -torsion lies in degree i and hence does not enter the present range. The top group is the reduction of the p -torsion-free module $\widehat{\Omega}_R^{i+p}$.

Since $G_i \in D^{\leq i}$, it does not alter the cohomology of the cofiber Q_i above degree i . The ring R is regular, so the regular-ring weak F -smoothness theorem [4, Theorem 4.15] gives $Q_i \in D^{\geq i+1}$. The map q_i identifies the degree- $i+p$ group with $K_{i+p}(R)$. If

$$F_i(R) = \mathrm{fib}(q_i : Q_i(R) \rightarrow K_{i+p}(R)),$$

then $F_i(R)$ has a single nonzero cohomology group, in degree $i+1$, canonically identified with Ω_R^i/p . Thus there is a canonical equivalence

$$F_i(R) \simeq K_i(R)[-1]. \tag{1}$$

The normalization in (1) is the Sen cokernel: $H^{i+1}F_i$ is the cokernel of multiplication by p on $\widehat{\Omega}_R^i$. In particular no splitting of the other terms of the Sen sequence is being chosen.

Now animate the entire diagram consisting of q_i , its specified nullhomotopy, its functorial fiber F_i , and the equivalence (1). After reduction modulo p , the functors G_n preserve sifted colimits by [2, Remark 5.5.10]; the K_n do so by construction of derived exterior powers, and finite fibers and cofibers preserve sifted colimits in a stable category. Hence left Kan extension from polynomial rings gives the coherent triangle for every animated ring. Rotating this animated fiber sequence defines β_i . This is stronger than identifying the objectwise maps on cohomology: the natural nullhomotopies and the maps of fibers are retained throughout.

We finally check compatibility with a polynomial variable before animation. The decompositions of Nygaard–Sen and Hodge terms for $R[X]$ are induced by the unit and by exterior multiplication with dX . Multiplicativity identifies θ on each summand with the corresponding multiplication by θ over R , and the specified nullhomotopy of its Hodge image is compatible with multiplication by dX . Therefore there is a canonical decomposition of maps

$$(Q_i(R[X]) \xrightarrow{q_i} K_{i+p}(R[X])) \simeq (Q_i(R) \xrightarrow{q_i} K_{i+p}(R))[X] \oplus (Q_{i-1}(R) \xrightarrow{q_{i-1}} K_{i+p-1}(R))[X][-1].$$

Taking functorial fibers gives the corresponding decomposition of $F_i(R[X])$. On the dX summand the Sen operator changes from Θ to $\Theta - 1$, so the cokernel in degree $i + 1$ is again the cokernel of multiplication by p , now on $\widehat{\Omega}_R^{i-1} dX$. Under the canonical identifications (1), the map on this summand is exterior multiplication by dX with coefficient one (up to the fixed convention for ordering differentials).

After shifting all three fiber terms to degree zero, they lie in the heart of the module derived category. Mapping spaces between such objects are discrete, so the just-identified maps of differential-form modules lift uniquely to the corresponding maps of complexes, together with their commuting squares. Animating this compatible diagram gives

$$\beta_i(R[X]) = \beta_i(R)[X] \oplus \beta_{i-1}(R)[X][-1],$$

as asserted. □

3.2 The Frobenius–Witt splitting

We use Mao’s Frobenius–Witt cotangent complex [13]. For a ring R , write FL_R for the pointed R/p -module of Frobenius–Witt differentials. It fits into the natural triangle

$$A_R \longrightarrow \mathrm{FL}_R \longrightarrow L_{R/\mathbf{Z}} \otimes_{R,F}^{\mathbf{L}} A_R \longrightarrow A_R[1]. \quad (2)$$

Mao proves that FL_V is a flat discrete V/p -module for every valuation ring V [13, Theorem A.2], and that $\mathrm{FL}_R = 0$ when the derived p -completion of R is perfectoid [13, Lemma A.1].

For a discrete ring, FL_R is generated by classes $w_R(x)$ with

$$\begin{aligned} w_R(xy) &= x^p w_R(y) + y^p w_R(x), \\ w_R(x + y) &= w_R(x) + w_R(y) - P(x, y) w_R(p), \\ w_R(1) &= 0, \end{aligned}$$

where $P(x, y) = ((x + y)^p - x^p - y^p)/p$; this is the universal arithmetic-extension description of [13, Section 2 and Theorem 3.6].

Lemma 3.2. *Assume that V is nondiscrete and put $A = V/p$, $M = \mathrm{FL}_V$. Then $w_V(p) = 0$ and*

$$F_A^* C_V \simeq A[1] \oplus M.$$

Consequently, for every $n \geq 0$,

$$F_A^* K_n(V) \simeq \bigoplus_{r=0}^n \bigwedge^r M[-r], \quad (3)$$

and every exterior power in this formula is flat.

Proof. Normalize $v(p) = 1$. Density gives $a \in V$ with $p^{-2} \leq v(a) < p^{-1}$ and put $b = p/a^p$. Then a, b lie in the maximal ideal and $a^{p^2} \in pV$. The Frobenius–Witt relations give

$$w_V(a^p) = pa^{p(p-1)}w_V(a) = 0, \quad w_V(p) = w_V(a^p b) = a^{p^2}w_V(b) = 0.$$

Thus the first map in (2) is zero. Since M is discrete, the zero-point splitting is unique and yields $F_A^* C_V \simeq A[1] \oplus M$. Derived exterior powers and the usual decalage formula give (3). Flatness of the exterior powers follows by expressing the flat module M as a filtered colimit of finite free modules. $\square$

3.3 The scalar calculation

We next show that, after Frobenius pullback, the map β_i of [theorem 3.1](#) is a unit on each common exterior-power summand.

Choose a, b as in the preceding proof and set

$$R_0 = (\mathbf{Z}_p[T, B]/(p - T^p B))_{(T, B)}, \quad A_0 = R_0/p.$$

Evaluation $T \mapsto a, B \mapsto b$ gives a local map $R_0 \rightarrow V$. The ring R_0 is regular local of dimension two and p -torsion-free.

Lemma 3.3. *For every $j \geq 0$, the Frobenius pullback of $\beta_j(R_0)$, after scalar extension $A_0 \rightarrow A$, acts on the canonical H^0 copy of A by a unit $u_j \in A^\times$.*

Proof. By regularity and [13, Theorem 5.2], FL_{R_0} is flat over A_0 . The Frobenius–Witt relation gives $w_{R_0}(p) = T^{p^2}w_{R_0}(B)$, which vanishes after base change to A because $a^{p^2} = 0$ in A . Hence the same splitting as in [theorem 3.2](#) produces a distinguished H^0 line, on which $F^*\beta_j$ acts by a scalar u_j .

Extend K to the completed algebraic closure and let W be its valuation ring. Then W is perfectoid and $A \rightarrow W/p$ is local. Mao’s perfectoid vanishing gives $\mathrm{FL}_W = 0$, while Bhatt–Mathew’s perfectoid Nygaard calculation gives $Q_j(W) = 0$ [4, Example 2.5]. Therefore $\beta_j(W)$ is an equivalence. Naturality shows that the image of u_j in W/p is a unit; because $A \rightarrow W/p$ reflects units, u_j was already a unit. $\square$

Lemma 3.4. *Under the splitting (3), the map*

$$H^r(F_A^* \beta_i(V)) : \bigwedge^r M \longrightarrow \bigwedge^r M, \quad 0 \leq r \leq i,$$

is multiplication by the unit $\pm u_{i-r}$.

Proof. For $x_1, \dots, x_r \in V$, evaluate

$$R_0[X_1, \dots, X_r] \longrightarrow V, \quad X_s \mapsto x_s.$$

The polynomial compatibility in [theorem 3.1](#) identifies the pure r -differential summand of β_i with $\beta_{i-r}(R_0)[-r]$. Under the Frobenius–Witt splitting, the generator maps to

$$w_V(x_1) \wedge \cdots \wedge w_V(x_r).$$

Hence the map is multiplication by $\pm u_{i-r}$ on these wedges. They generate $\bigwedge^r M$, proving the assertion. The sign is the fixed Koszul sign coming from the order of the polynomial differentials; it is a unit also when $p = 2$. $\square$

3.4 Frobenius pullback reflects Tor bounds

Lemma 3.5. *Let $A = V/p$ for a rank-one mixed-characteristic valuation ring. Derived Frobenius pullback F_A^* reflects Tor-amplitude intervals and one-sided Tor lower bounds.*

Proof. Let W be the valuation ring in the completed algebraic closure of K and put $D = W/p$. The map $A \rightarrow D$ is faithfully flat. Frobenius on D factors as a nilpotent quotient

$$D \longrightarrow D/(\varpi) \xrightarrow{\sim} D, \quad \varpi^p = p.$$

Base change along a nilpotent quotient reflects Tor bounds: if $J^N = 0$, every discrete module has a finite filtration with R/J -module subquotients. Thus F_D^* reflects Tor bounds. Frobenius naturality gives

$$(F_A^* E) \otimes_A^{\mathbf{L}} D \simeq F_D^*(E \otimes_A^{\mathbf{L}} D).$$

A bound on $F_A^* E$ therefore gives the same bound on $E \otimes_A^{\mathbf{L}} D$, and faithful flatness reflects it back to E . $\square$

3.5 Weak F -smoothness

Theorem 3.6. *Let V be a p -complete rank-one valuation ring of mixed characteristic. Then V is p -quasisyntomic and weakly F -smooth in the sense of [\[4, Definition 4.2\]](#). If V is nondiscrete, then for every $i \geq 0$*

$$F_A^* Q_i(V) \simeq \bigoplus_{r=i+1}^{i+p} \bigwedge_A^r M[-r], \quad A = V/p, \quad M = \mathrm{FL}_V. \quad (4)$$

In particular $Q_i(V)$ has A -Tor amplitude bounded below by $i + 1$.

Moreover, for every twist $n \in \mathbf{Z}$ and $m \geq 0$,

$$\mathrm{fib}(\theta : \mathrm{gr}_N^m \Delta_V \{n\}/p \rightarrow \mathrm{gr}_N^{m+p} \Delta_V \{n+p-1\}/p) \in \mathrm{D}^{\geq m+2}, \quad (5)$$

$$\mathrm{fib}(\varphi_m : \mathrm{gr}_N^m \Delta_V \{n\}/p \rightarrow \overline{\Delta}_V \{m\}/p) \in \mathrm{D}^{\geq m+2}. \quad (6)$$

Proof. Assume first that V is nondiscrete. Apply F_A^* to the triangle of [theorem 3.1](#). By [theorem 3.2](#), decompose

$$F_A^* K_{i+p}(V) = L \oplus H,$$

where

$$L = \bigoplus_{r=0}^i \bigwedge_A^r M[-r], \quad H = \bigoplus_{r=i+1}^{i+p} \bigwedge_A^r M[-r].$$

By [theorem 3.4](#), the restriction of $F_A^* \beta_i$ to L is an equivalence. An elementary change of coordinates on $L \oplus H$ removes the off-diagonal map from H to the target, so the fiber is H . This proves (4). Since every $\bigwedge^r M$ is flat, the lower Tor bound is $i + 1$, and [theorem 3.5](#) reflects it to $Q_i(V)$.

The same reflection argument applied to $F_A^* C_V = A[1] \oplus M$ shows that $L_{A/\mathbf{F}_p}$ has Tor amplitude $[-1, 0]$. Since V is p -complete and p -torsion-free, this is the required cotangent condition for p -quasisyntomicity.

The fiber of $\theta : G_i \rightarrow G_{i+p}$ is $Q_i[-1]$. Changing rings from A to V loses at most one Tor degree because p is a nonzerodivisor, giving exactly the weak F -smoothness bound of [4, Definition 4.2]. In particular $Q_i(V) \in D^{\geq i+1}$, so the untwisted theta fiber is in $D^{\geq i+2}$.

For the twisted statement, the associated graded Breuil–Kisin line has a canonical degree-zero unit: multiplication identifies every twisted graded piece with the untwisted one; see [2, Corollary 5.5.14 and Remark 5.5.15]. The graded image of the syntomic Bott lift is θ times this unit [4, Construction 2.8 and Proposition 2.9]. Hence (5) is the untwisted bound transported through these equivalences.

Finally, divided Frobenius is the telescope obtained by inverting θ on the Nygaard graded algebra [4, Proposition 2.6]. The successive cofibers have lower bounds tending to infinity, so the fiber from the first term to the telescope lies in $D^{\geq m+2}$. This gives (6).

If V is discrete, it is a complete DVR and therefore regular noetherian. The same assertions follow from F -smoothness of regular rings [4, Theorem 4.15], together with the same graded twist and telescope arguments. $\square$

4 Bott strictness and Nygaard-completion defects

We now isolate the exact discrepancy between Bott multiplication on the uncompleted and completed prismatic terms. The output is a family of twist-indexed completion defects $\mathcal{E}_k$ and a truncated comparison between those defects and Bott cofibers in syntomic cohomology.

4.1 Finite Bott strictness on QRSP rings

Let R be a p -torsion-free quasiregular semiperfectoid (QRSP) ring. Write

$$D = \Delta_R/p, \quad N^r = N^{\geq r} \Delta_R/p, \quad L = \Delta_R\{1\}/p.$$

Locally choose a generator d of the prism ideal and a generator e of the Breuil–Kisin line, normalized by

$$\varphi_1(e) = \frac{u}{d}e, \quad u \in D^\times.$$

In this trivialization the mod- p Bott class has the form

$$b = ce^{p-1}, \quad c = d/u,$$

up to a nonzero scalar in $\mathbf{F}_p$; cf. [4, Constructions 2.7–2.8, Proposition 2.9 and Remark 2.10].

Lemma 4.1 (Finite Bott strictness). *Multiplication by c is injective on D , and for every $r \geq 0$,*

$$cD \cap N^{r+p} = cN^r. \tag{7}$$

Consequently, for every $q \geq p$ there is a short exact sequence

$$0 \longrightarrow D/N^{q-p} \xrightarrow{c} D/N^q \longrightarrow D/(cD + N^q) \longrightarrow 0. \tag{8}$$

The assertions are intrinsic after restoring the Breuil–Kisin twists.

Proof. For a p -torsion-free QRSP ring, divided Frobenius identifies the Nygaard graded pieces with the relative conjugate filtration, and these pieces inject into D after reduction modulo p ; see [7, Theorem 12.2 and Proposition 12.8]. Moreover (p, d) is a regular sequence on the prismatic ring, so c is a nonzerodivisor modulo p . Since

$$\frac{\varphi(c)}{d^p} = u^{-p} \in D^\times,$$

we have $c \in N^p$, and multiplicativity of divided Frobenius shows that

$$c : \mathrm{gr}_N^r D \longrightarrow \mathrm{gr}_N^{r+p} D$$

is injective. If $cy \in N^{r+p}$, begin with $y \in N^0 = D$ and induct on $s < r$: if $y \in N^s$, then the image of cy in gr_N^{s+p} vanishes, hence the injectivity just proved forces $y \in N^{s+1}$. Thus $y \in N^r$, proving (7). The quotient sequence follows immediately. $\square$

4.2 The completion defect

For a twist k set

$$H_k(R) = \overline{\Delta}_R\{k\}/p.$$

For $q \geq p$, let $B_q^k(R)$ be the intrinsic finite quotient defined as the cofiber of Bott multiplication

$$(\Delta_R\{k-p+1\}/p)/N^{q-p} \xrightarrow{v_1} (\Delta_R\{k\}/p)/N^q.$$

On a QRSP chart, [theorem 4.1](#) identifies this with the ordinary quotient

$$D\{k\}/(ID\{k\} + N^q D\{k\}).$$

For general quasisyntomic R , this finite cofiber is the definition; in particular it is compatible with quasisyntomic descent and with filtered colimits modulo p .

Define the induced filtration on H_k by

$$\mathcal{F}^q H^0(H_k) = \ker(H^0 H_k \rightarrow H^0 B_q^k),$$

and let

$$\widehat{H}_k = \mathrm{Rlim}_q B_q^k, \quad \mathcal{E}_k(R) = \mathrm{cofib}(H_k(R) \rightarrow \widehat{H}_k(R)).$$

Lemma 4.2 (Bott fiber versus completion defect). *Let $j \in \mathbf{Z}$, put $k = j + p - 1$, and define*

$$Y_j = \mathrm{fib}(b : \Delta_R\{j\}/p \rightarrow \Delta_R\{k\}/p),$$

$$\widehat{Y}_j = \mathrm{fib}(b : \widehat{\Delta}_R\{j\}/p \rightarrow \widehat{\Delta}_R\{k\}/p).$$

Then

$$Y_j \simeq H_k[-1], \quad \widehat{Y}_j \simeq \widehat{H}_k[-1],$$

and therefore

$$\mathrm{fib}(Y_j \rightarrow \widehat{Y}_j) \simeq \mathcal{E}_k[-2]. \tag{9}$$

On a p -torsion-free QRSP ring, H_k and $\widehat{H}_k$ are discrete and $\mathcal{E}_k$ has amplitude contained in $[-1, 0]$.

Proof. On a QRSP chart, multiplication by b is multiplication by the nonzerodivisor c after trivializing the twist. Its cokernel is the Hodge–Tate quotient H_k , giving the first equivalence. Taking inverse limits in (8) is exact because all transition maps in the quotient towers are surjective, and yields the second equivalence. Reduction modulo p commutes with these derived limits, since $\mathbf{Z}/p$ is perfect over $\mathbf{Z}$. Quasisyntomic descent globalizes the identifications. Equation (9) is then the fiber of the map between the two Bott fibers. $\square$

4.3 From graded bounds to completed Bott fibers

The weak F -smoothness bounds of [theorem 3.6](#) remain valid after Nygaard completion in the precise form needed below.

Lemma 4.3. *For $m \geq 0$ and every twist n ,*

$$\mathrm{fib}\left(v_1 : N^{\geq m} \widehat{\Delta}_V\{n\}/p \rightarrow N^{\geq m+p} \widehat{\Delta}_V\{n+p-1\}/p\right) \in \mathrm{D}^{\geq m+2}. \quad (10)$$

Moreover divided Frobenius factors naturally as

$$N^{\geq n} \widehat{\Delta}_V\{n\}/p \xrightarrow{\widetilde{\varphi}_n} H_n(V) \longrightarrow \widehat{H}_n(V),$$

with

$$\mathrm{fib}(\widetilde{\varphi}_n) \in \mathrm{D}^{\geq n+2} \quad (n \geq 0). \quad (11)$$

Proof. The completed Nygaard filtration is complete by construction. The graded pieces of the fiber in (10) are the twisted theta fibers of (5), with indices at least m . Finite extensions and inverse limits preserve the common lower bound $m+2$, which proves (10).

For divided Frobenius, the associated graded map is the map to the theta-localized Hodge–Tate complex. Equation (6) gives the lower bound on every graded fiber. The filtered factorization through the completed source is the canonical factorization of [2, Remarks 5.7.8 and 5.7.10]. Passing to the complete filtered objects gives (11). $\square$

Let

$$C_j(V) = \mathrm{cofib}(v_1 : S_j(V) \rightarrow S_{j+p-1}(V)).$$

The following truncated identification is the exact bridge between Bott cofibers and completion defects. We give it here because it will be used twice: first to obtain the initial defect vanishing and finally in the induction of [section 9](#).

Proposition 4.4 (Truncated Bott–defect comparison). *Let $j \geq 1$ and $k = j + p - 1$. Then*

$$\tau^{\leq j-1} C_j(V) \simeq \tau^{\leq j-1} (\mathcal{E}_k(V)[-1]). \quad (12)$$

Equivalently, for $1 \leq q \leq j$,

$$H^{q-1} C_j(V) \simeq H^{q-2} \mathcal{E}_k(V).$$

Proof. Syntomic cohomology can be computed on Nygaard-completed terms:

$$S_i(V) \simeq \mathrm{fib}(\widehat{\varphi}_i - \widehat{\mathrm{can}} : N^{\geq i} \widehat{\Delta}_V\{i\}/p \rightarrow \widehat{\Delta}_V\{i\}/p)$$

by [2, Proposition 7.4.6]. Taking the fiber of Bott multiplication between the two syntomic squares gives

$$F_j := \mathrm{fib}(S_j \rightarrow S_k) = \mathrm{fib}(\alpha - \beta : X_j \rightarrow \widehat{Y}_j),$$

where X_j is the Bott fiber on the completed Nygaard sources. The divided-Frobenius map factors

$$X_j \xrightarrow{f} Y_j \xrightarrow{g} \widehat{Y}_j, \quad \alpha = gf,$$

and (11) implies $\mathrm{fib}(f) \in \mathrm{D}^{\geq j+2}$. The canonical-map contribution β factors through

$$Z_j = \mathrm{fib}\left(v_1 : N^{\geq j-1} \widehat{\Delta}_V\{j\}/p \rightarrow N^{\geq j+p-1} \widehat{\Delta}_V\{k\}/p\right),$$

which belongs to $D^{\geq j+1}$ by (10).

Apply $\tau^{\leq j}$. Then f becomes an equivalence and the factorization through Z_j makes β canonically zero in $D^{\leq j}$. Therefore

$$\tau^{\leq j} F_j \simeq \tau^{\leq j} \text{fib}(g).$$

By (9), $\text{fib}(g) = \mathcal{E}_k[-2]$. Since $C_j = F_j[1]$, shifting gives (12). $\square$

5 Separation on the first perfectoid overlap

The goal is to prove the initial estimate $\mathcal{E}_k(V) \in D^{\geq 1}$. It is enough to establish separation of the induced Nygaard filtration on the first nontrivial term of a perfectoid Čech nerve. The point of the next two subsections is to make the required separation estimate uniform in the conjugate degree whenever the number of complete-intersection equations is fixed.

5.1 A universal separation estimate

Let $W = \mathcal{O}_C$ for an algebraically closed perfectoid field of mixed characteristic, and consider

$$S_r = W\langle X_1^{1/p^\infty}, \dots, X_r^{1/p^\infty} \rangle / (X_1, \dots, X_r).$$

Give $\overline{\Delta}_{S_r}/p$ the relative conjugate filtration over W , and let the induced Nygaard filtration be the image of the prismatic Nygaard steps.

Lemma 5.1 (Universal bound). *For every $s \geq 0$,*

$$\mathcal{N}^{\geq n}(\overline{\Delta}_{S_r}/p) \cap \text{Fil}_s^{\text{conj}}(\overline{\Delta}_{S_r}/p) = 0 \quad \text{if } n \geq ps + r(p-1) + p. \quad (13)$$

In particular the induced Nygaard filtration is separated.

Proof. The explicit QRSP calculation of [7, Lemmas 12.4 and 12.7] gives a basis indexed by $\mathbb{N}[1/p]^r$. In one variable the basis element e_u has Nygaard coefficient

$$t^{\max(n - \lfloor u \rfloor, 0)},$$

while the relative conjugate degree is $\lfloor u/p \rfloor$. Tensoring the calculation gives the multivariable basis. If a basis vector has total conjugate degree at most s , then

$$\sum_j \lfloor u_j \rfloor \leq p \sum_j \lfloor u_j/p \rfloor + r(p-1) \leq ps + r(p-1).$$

The stated bound forces its Nygaard coefficient to be divisible by t^p ; on the Hodge–Tate specialization modulo p one has $t^p = 0$. This proves the intersection statement. Exhaustiveness of the conjugate filtration gives separatedness. $\square$

Lemma 5.2 (Uniform complete-intersection separation). *Let P be a p -torsion-free perfectoid W -algebra and let $f_1, \dots, f_r \in P$ be a sequence whose images form a Koszul-regular sequence on P/p . Put*

$$S = (P/(f_1, \dots, f_r))_p^\wedge$$

and assume S is p -torsion-free. Then the image of $H^0(N^{\geq n} \Delta_S \otimes^{\mathbf{L}} \mathbf{F}_p)$ in $\overline{\Delta}_S/p$ has zero intersection with $\text{Fil}_s^{\text{conj}}$ whenever

$$n \geq ps + r(p-1) + p.$$

The same conclusion holds after filtered colimits of such presentations with a uniform bound on r .

Proof. We first treat the case in which compatible p -power roots of the f_i have been chosen in P . Put

$$P_0 = W\langle X_1^{1/p^\infty}, \dots, X_r^{1/p^\infty} \rangle, \quad S_0 = P_0/(X_1, \dots, X_r),$$

and map P_0 to P by sending X_i^{1/p^m} to the chosen roots of f_i . Since the images of the f_i form a Koszul-regular sequence on P/p , the derived base change

$$S_0/p \otimes_{P_0/p}^{\mathbf{L}} P/p \simeq S/p \quad (14)$$

is discrete.

We use the perfectoid base-change statement contained in the proof of [7, Proposition 12.8]: prismatic cohomology and the relative conjugate filtration commute with derived p -complete base change along perfectoid algebras, and the transported good filtration is the Nygaard filtration. Thus

$$N^{\geq n} \Delta_S \simeq N^{\geq n} \Delta_{S_0} \widehat{\otimes}_{\Delta_{S_0}}^{\mathbf{L}} \Delta_S. \quad (15)$$

It is important that we use this statement at the level of the filtered complexes, rather than merely an isomorphism of associated graded objects.

We next make the universal factorization explicit. In the basis of [theorem 5.1](#), write each multi-index uniquely as

$$\mathbf{u} = \mathbf{a} + p\mathbf{k}, \quad 0 \leq a_i < p, \quad \mathbf{k} \in \mathbf{N}^r,$$

and set $g_{\mathbf{k}} = e_{p\mathbf{k}}$. Multiplication by the finitely many $e_{\mathbf{a}}$ identifies the Hodge–Tate specialization as

$$\overline{\Delta}_{S_0}/p = \bigoplus_{\mathbf{k} \in \mathbf{N}^r} (S_0/p)g_{\mathbf{k}}, \quad \text{Fil}_s^{\text{conj}} = \bigoplus_{|\mathbf{k}| \leq s} (S_0/p)g_{\mathbf{k}}. \quad (16)$$

The coefficient occurring in $e_{\mathbf{a}}g_{\mathbf{k}} = u_{\mathbf{a},\mathbf{k}}e_{\mathbf{a}+p\mathbf{k}}$ is a unit: all new q -factorial factors have indices strictly between consecutive multiples of p . Hence (16) is an actual direct-sum decomposition over S_0/p , not merely a splitting of associated graded modules. Write $H_{>s}$ for the complementary sum indexed by $|\mathbf{k}| > s$.

There is a derived point that cannot be omitted. For a general QRSP ring, reducing a Nygaard step modulo the prism ideal may create negative cohomology, so the intersection statement cannot be transported merely from H^0 of the universal source. In the universal model, however, the explicit formula of [7, Lemma 12.7] gives

$$N^{\geq n} \Delta_{S_0} = \widehat{\bigoplus_{\mathbf{u}}} t^{\max(n - \sum_i \lfloor u_i \rfloor, 0)} A_{\text{inf}}(W) e_{\mathbf{u}},$$

where $t = \varphi^{-1}(\xi)$ and ξ generates the prism ideal of W . Each coefficient ideal is abstractly free of rank one because t is a nonzerodivisor. Since (p, ξ) is a regular sequence on $A_{\text{inf}}(W)$, the Nygaard step is (p, ξ) -completely flat over $A_{\text{inf}}(W)$. Therefore

$$D_n := N^{\geq n} \Delta_{S_0} \otimes_{A_{\text{inf}}(W)}^{\mathbf{L}} A_{\text{inf}}(W)/(p, \xi) \quad (17)$$

is discrete.

The numerical estimate of [theorem 5.1](#) now says more than that the image of $H^0(D_n)$ avoids the low conjugate part: the entire morphism of discrete P_0/p -modules

$$D_n \longrightarrow \overline{\Delta}_{S_0}/p$$

factors through the complementary summand $H_{>s}$ whenever $n \geq ps + r(p-1) + p$. Tensor this factorization over P_0/p with P/p . By (14), the low and high sums in (16) remain discrete complementary free S/p -modules after derived base change. Equations (15) and (17) identify the transported source map with the Nygaard map for S after Hodge–Tate reduction. Although that transported source may now possess negative cohomology, its whole morphism still factors through the discrete high summand. Consequently the image on H^0 cannot meet the low conjugate summand. This proves the desired intersection estimate when the roots have been chosen.

We remove the choice of roots by Andre’s flatness lemma [7, Theorem 7.14]: after a p -completely faithfully flat perfectoid extension $P \rightarrow P'$, every f_i admits compatible p -power roots. Regularity modulo p is preserved by this flat base change, and

$$S' = (P'/(f_1, \dots, f_r))_p^\wedge$$

is a p -completely faithfully flat quasisyntomic extension of S . The conjugate filtration and its Hodge–Tate graded terms base change along $S/p \rightarrow S'/p$, so $H^0(\overline{\Delta}_S/p) \rightarrow H^0(\overline{\Delta}_{S'}/p)$ is injective. Naturality carries the intersection in question to the zero intersection over S' , proving the result over S .

Finally suppose S is obtained as a filtered colimit of such complete intersections with a uniform number r of equations. The finite Nygaard quotients and finite conjugate pieces commute with filtered colimits: both are built from finitely many Nygaard graded pieces, and those are the corresponding finite conjugate pieces by divided Frobenius. Hence a class in a fixed conjugate step and the equality witnessing its membership in a fixed Nygaard level both occur at a common finite stage. Applying the same uniform estimate at that stage proves the filtered-colimit assertion. $\square$

5.2 The first overlap

Let C be the completion of an algebraic closure of K , let $W = \mathcal{O}_C$, and write

$$R = (W \otimes_V W)_p^\wedge.$$

We prove that the induced Nygaard filtration on $\overline{\Delta}_R/p$ is separated.

We use the valuation-theoretic description of tame and degree- p extensions from Gabber–Ramero [8, Proposition 6.2.12, Corollary 6.2.14, Proposition 6.3.11]. In particular, after passing to the maximal tame extension, every finite Galois extension is built from a finite tower of degree- p extensions, and the integral closure at each degree- p step is a filtered union of monogenic finite-free hypersurfaces.

Lemma 5.3. *Let V^t be the valuation ring in the maximal tame extension K^t/K . Then*

$$B = (W \otimes_V V^t)_p^\wedge$$

is p -torsion-free perfectoid.

Proof. The tensor product is flat over W , hence p -torsion-free. The relative Frobenius

$$(V^t/p) \otimes_{V/p, F}^{\mathbf{L}} V/p \longrightarrow V^t/p$$

is an isomorphism. Indeed, after strict henselization, a finite tame cyclic extension of prime degree $\ell \neq p$ has a basis with distinct value cosets. Each nontrivial coset lattice is a directed union of principal lattices with values approaching zero; Frobenius carries this directed system

isomorphically to the corresponding system in the p -multiple value coset. Passing to the filtered union gives relative perfection.

Base change to W/p therefore identifies Frobenius $B/\pi \rightarrow B/p$ for $\pi = p^{1/p} \in W$ with an isomorphism. Since π is a nonzerodivisor, the perfectoid criterion of [5, Lemma 3.10] implies that B is perfectoid. $\square$

Lemma 5.4 (Triangular finite models). *Let*

$$K_0 \subset K_1 \subset \cdots \subset K_r$$

be a tower of henselian rank-one valued fields such that each K_i is tamely closed and $[K_i : K_{i-1}] = p$. Write $A_i = K_i^+$. Then A_r is the filtered union of A_0 -subalgebras admitting triangular presentations

$$A_0[X_1, \dots, X_r]/(g_1, \dots, g_r), \quad (18)$$

where $g_i \in A_0[X_1, \dots, X_i]$ is monic of degree p in X_i . Every such algebra is finite free of rank p^r over A_0 . After any p -torsion-free perfectoid base change $A_0 \rightarrow B$, the completed base changes are complete intersections with the same r equations, regular modulo p .

Proof. For one degree- p extension, the assertion is exactly the filtered monogenic description of [8, Proposition 6.3.11]: if $K = K^t$ and $[E : K] = p$, then E^+ is the filtered union of finite K^+ -algebras $K^+[x]$ generated by integral primitive elements. Their minimal polynomials are monic of degree p .

We prove the cofinal triangular description by induction on r . Let $F \subset A_r$ be a finite set. By the degree- p statement applied to K_r/K_{r-1} , choose an integral primitive element $x_r \in A_r$ such that

$$F \subset A_{r-1}[x_r].$$

Only finitely many coefficients occur in the monic minimal polynomial of x_r and in fixed polynomial expressions of the elements of F in terms of x_r . By the induction hypothesis, all those coefficients lie in a single triangular subalgebra

$$D_{r-1} = A_0[x_1, \dots, x_{r-1}] \subset A_{r-1}.$$

Its fraction field is K_{r-1} . The minimal polynomial g_r of x_r therefore lies in $D_{r-1}[X_r]$, is monic of degree p , and the natural map

$$D_r := D_{r-1}[X_r]/(g_r) \longrightarrow A_r$$

is injective: D_r is free of rank p over D_{r-1} , and after inverting the nonzero elements of D_{r-1} its generic fiber is the field K_r . By construction D_r contains F .

Thus every finite subset of A_r lies in a model of the form (18). Given two such models, apply the same construction to the union of their finite sets of algebra generators to obtain a third containing both; the family is therefore filtered by inclusion. Successively using the monic bases $1, X_i, \dots, X_i^{p-1}$ shows that every model is finite free of rank p^r over A_0 .

Choose polynomial representatives for the coefficients of each g_i in the preceding variables. Because g_i is monic in the new variable X_i , it remains a nonzerodivisor after quotienting by $g_1, \dots, g_{i-1}$ and after arbitrary base change. Hence $(g_1, \dots, g_r)$ is a regular sequence, also after reduction modulo p whenever the resulting algebra is p -torsion-free. The same monic bases show that the completed perfectoid base changes remain finite free and give complete intersections with exactly r equations.

In the application below the intermediate fields remain tamely closed. Indeed, for every separable E/K one has $E^t = EK^t$ [8, Section 6.2.18]; if $K = K^t$ and E/K occurs in a finite p -tower over K , this identity gives $E = E^t$. $\square$

Proposition 5.5 (First-overlap separation). *The induced Nygaard filtration on $H^0(\overline{\Delta}_R/p)$ is separated.*

Proof. Let L/K range over finite subextensions of the chosen algebraic closure and put

$$T_L = (W \otimes_V V_L)_p^\wedge.$$

Every Hodge–Tate class on R comes from some T_L , because $R/p = \operatorname{colim}_L T_L/p$ and the finite conjugate filtration commutes with this colimit.

After replacing L by a finite Galois extension, set $E = LK^t$. The extension E/K^t is a finite p -extension. Choose a subnormal tower

$$K^t = F_0 \subset F_1 \subset \cdots \subset F_r = E, \quad [F_i : F_{i-1}] = p.$$

Every intermediate field F_i is again tamely closed, and [theorem 5.4](#) gives a cofinal filtered system of subalgebras of E^+ with triangular presentations

$$V^t[X_1, \dots, X_r]/(g_1, \dots, g_r),$$

where each g_i is monic of degree p in the new variable. After base change along $V^t \rightarrow B$ and p -completion, these are p -torsion-free complete intersections with exactly r equations, regular modulo p . The point of the lemma is that the same integer r works for the whole filtered system attached to this fixed finite extension E/K^t .

The map

$$T_L \longrightarrow (B \otimes_{V^t} E^+)_p^\wedge$$

is p -completely faithfully flat and quasisyntomic: valuation extensions are flat, and the cotangent complex has the valuation-theoretic Tor bound of [\[8, Theorem 6.5.8\]](#). The target is a filtered colimit of the complete intersections just described. By [theorem 5.2](#), its Nygaard filtration is separated, with a bound depending on the fixed tower length r and on the conjugate degree of the chosen class. Quasisyntomic descent for the finite quotients reflects this separation to T_L .

It remains to descend the conclusion without changing the finite stage. For any p -completely faithfully flat quasisyntomic map $T \rightarrow U$, both $\overline{\Delta}_T/p$ and every finite quotient $B_q(T)$ are connective: this follows from the Hodge–Tate filtration and the quasisyntomic Tor bound. Čech descent therefore makes

$$H^0(\overline{\Delta}_T/p) \longrightarrow H^0(\overline{\Delta}_U/p), \quad H^0 B_q(T) \longrightarrow H^0 B_q(U)$$

injective. Hence the induced filtration is strict on H^0 :

$$H^0(\overline{\Delta}_T/p) \cap \mathcal{F}^q H^0(\overline{\Delta}_U/p) = \mathcal{F}^q H^0(\overline{\Delta}_T/p). \quad (19)$$

Now let $x \in H^0(\overline{\Delta}_R/p)$ lie in every Nygaard level and choose L so that it comes from $x_L \in H^0(\overline{\Delta}_{T_L}/p)$. The map $T_L \rightarrow R$ is a p -completely faithfully flat quasisyntomic valuation base change. Applying (19) for every q shows that the *same* class x_L lies in every Nygaard level of T_L . Separatedness on T_L gives $x_L = 0$, hence $x = 0$. $\square$

5.3 The initial defect bound

Let W be the perfectoid valuation cover of V and

$$A^a = (W^{\otimes_V (a+1)})_p^\wedge$$

its Čech nerve. By [theorem 2.3](#), these rings are p -torsion-free QRSP, and prismatic, finite Nygaard, Hodge–Tate and syntomic cohomology satisfy quasisyntomic descent on this nerve.

Proposition 5.6 (Initial connectivity of the defects). *If V is nondiscrete, then for every $k \in \mathbf{Z}$,*

$$\mathcal{E}_k(V) \in D^{\geq 1}.$$

Proof. On each QRSP term, [theorem 4.2](#) gives $\mathcal{E}_k(A^a) \in D^{[-1,0]}$. For $a = 0$ the ring is perfectoid, so the induced filtration is already complete and $\mathcal{E}_k(A^0) = 0$. By [theorem 5.5](#), the induced filtration on A^1 is separated, hence

$$H^{-1}\mathcal{E}_k(A^1) = 0.$$

Descent gives

$$\mathcal{E}_k(V) \simeq \text{Tot}(\mathcal{E}_k(A^\bullet)).$$

The descent spectral sequence

$$E_1^{a,b} = H^b \mathcal{E}_k(A^a) \implies H^{a+b} \mathcal{E}_k(V)$$

has only rows $b = -1, 0$. In total degree 0, the only possible terms are $(0, 0)$ and $(1, -1)$, and both vanish by the preceding observations. In total degree -1 only $(0, -1)$ can occur, and it also vanishes. Hence $\mathcal{E}_k(V) \in D^{\geq 1}$. $\square$

6 The Milnor endpoint and the valuation double

The second independent input to the final induction is injectivity in the endpoint degree. The argument is purely algebraic after the syntomic excision statement below. We first construct the degree-two and degree-three operations that will later be propagated to all degrees.

6.1 Milnor and Dennis–Stein conventions

For a local ring R , let $K_*^M(R)$ denote ordinary Milnor K -theory and $\widehat{K}_*^M(R)$ Kerz’s improved Milnor K -theory. For every fixed m there is a constant M_m , independent of R , such that

$$|R/\mathfrak{m}_R| > M_m \implies K_m^M(R) \xrightarrow{\sim} \widehat{K}_m^M(R); \quad (20)$$

see [\[11, Proposition 10\(5\)\]](#). We will always enlarge the residue field only after fixing the finite range of degrees in which this comparison is needed.

We also use the absolute Dennis–Stein presentation of K_2 . If a commutative ring is 3-fold stable, then $K_2(R)$ is generated by symbols $\langle a, b \rangle$, defined when $1 + ab \in R^\times$, with relations

$$\langle a, b \rangle = -\langle -b, -a \rangle, \quad (21)$$

$$\langle a, b \rangle + \langle a, c \rangle = \langle a, b + c + abc \rangle, \quad (22)$$

$$\langle a, bc \rangle = \langle ab, c \rangle + \langle ac, b \rangle. \quad (23)$$

This is van der Kallen’s theorem [\[17, Theorem 7.1\]](#). If R is local with more than five residue elements, then it is 5-fold stable and the natural symbol map

$$K_2^M(R) \xrightarrow{\sim} K_2(R) \quad (24)$$

is an isomorphism [\[17, Theorem 8.4 and Corollary 8.5\]](#). In particular

$$\{u, v\} = -\{v, u\}, \quad \{u, -u\} = 0, \quad \{u, u\} = \{u, -1\} \quad (25)$$

for units u, v . These are integral identities, so the last formula remains valid modulo 2; the class $\{-1\}$ must then be retained.

Fix for the rest of this section a nondiscrete, p -complete rank-one valuation ring V with maximal ideal $\mathfrak{m}$, residue field k , and fraction field K . After fixing a finite degree range, we will first assume that k is large enough for all comparisons in that range. Put

$$A_m = K_m^M(V)/p, \quad G = K^\times/K^{\times p}, \quad P = \ker(A_1 \rightarrow k^\times/k^{\times p}), \quad J = PA \subset A = \bigoplus_{m \geq 0} A_m, \quad (26)$$

and write $e = \{-1\} \in A_1$. Every element of P is represented by a principal unit. By [theorem 2.2](#), every $g \in G$ has representatives of arbitrarily small positive valuation.

6.2 The valuation double and excision

Set

$$T = V \times_k V$$

with projections $q_0, q_1 : T \rightarrow V$ and diagonal section $s : V \rightarrow T$.

Lemma 6.1 (Tor-unitality of the double). *The square*

$$\begin{array}{ccc} T & \xrightarrow{q_0} & V \\ q_1 \downarrow & & \downarrow \\ V & \longrightarrow & k \end{array}$$

is a Tor-unital Milnor square. Consequently the induced square of nonconnective algebraic K -theory spectra is homotopy cartesian and

$$\ker(K_2(T) \xrightarrow{q_0} K_2(V)) \simeq K_2(V, \mathfrak{m}), \quad (27)$$

compatibly with the map to $K_2(V)$ induced by q_1 .

Proof. Nondiscreteness implies $\mathfrak{m}^2 = \mathfrak{m}$: if $0 \neq a \in \mathfrak{m}$, choose b with $0 < v(b) < v(a)$ and write $a = b(a/b)$. Every ideal in a valuation domain is a filtered union of principal ideals, hence $\mathfrak{m}$ is flat. The resolution

$$0 \rightarrow \mathfrak{m} \rightarrow V \rightarrow k \rightarrow 0$$

therefore gives $\mathrm{Tor}_i^V(k, k) = 0$ for $i > 0$.

Let $I = 0 \times \mathfrak{m} \subset T$. Under q_1 it maps isomorphically to $\mathfrak{m}$, while $T/I \simeq V$ through q_0 . Tor-unitality is independent of the ambient ring for an ideal carried isomorphically to an ideal [[14](#), Remark 1.9]; hence $q_0 : T \rightarrow V$ is Tor-unital. Tamme's excision theorem [[15](#), Theorem 2] makes the K -theory square cartesian. Since q_0 is split by s , taking π_2 identifies its fiber with the relative group in ([27](#)). $\square$

The same square is excisive for syntomic cohomology. This is important because it gives all-degree Milnor excision without requiring a relative presentation of Milnor K -theory.

Proposition 6.2 (Syntomic excision on the double). *For every $n \in \mathbf{Z}$ and $r \geq 1$, the square*

$$\begin{array}{ccc} \mathbf{Z}/p^r(n)_{\mathrm{syn}}(T) & \longrightarrow & \mathbf{Z}/p^r(n)_{\mathrm{syn}}(V) \\ \downarrow & & \downarrow \\ \mathbf{Z}/p^r(n)_{\mathrm{syn}}(V) & \longrightarrow & \mathbf{Z}/p^r(n)_{\mathrm{syn}}(k) \end{array}$$

is homotopy cartesian.

Proof. A homological epimorphism $R \rightarrow S$ satisfies $L_{S/R} = 0$: indeed $S \otimes_R^{\mathbf{L}} S \simeq S$, and cotangent base change followed by the transitivity triangle gives the claim. Thus, for every i , the four cotangent exterior powers

$$L \bigwedge^i L_{-/Z}$$

form a cartesian square on the valuation double. Equivalently, for every T -complex M there is a natural fiber sequence

$$M \longrightarrow (M \otimes_T^{\mathbf{L}} V) \oplus (M \otimes_T^{\mathbf{L}} V) \longrightarrow M \otimes_T^{\mathbf{L}} k,$$

obtained by tensoring $0 \rightarrow T \rightarrow V \oplus V \rightarrow k \rightarrow 0$.

The diffracted Hodge complex has its functorial conjugate filtration with graded pieces

$$(L \bigwedge^i L_{R/Z})_p^\wedge[-i],$$

and the Nygaard graded pieces are fibers of the corresponding Sen maps; see [2, Construction 4.7.1 and Remark 5.5.8]. Induction on finite conjugate steps therefore gives cartesian squares of every Nygaard graded piece. Finite Nygaard quotients are finite extensions of these graded pieces, so they are cartesian as well. Passing to the inverse limit gives cartesian squares of all Nygaard-completed steps. Finally, syntomic cohomology is the fiber of divided Frobenius minus the canonical map on these completed terms [2, Proposition 7.4.6]. Fibers preserve cartesian squares, proving the assertion. Derived reduction modulo p^r is exact and therefore preserves the result. $\square$

Corollary 6.3 (Milnor excision). *For every $n \geq 0$ and $r \geq 1$,*

$$\widehat{K}_n^{\mathbf{M}}(T)/p^r \xrightarrow{\sim} \widehat{K}_n^{\mathbf{M}}(V)/p^r \times_{\widehat{K}_n^{\mathbf{M}}(k)/p^r} \widehat{K}_n^{\mathbf{M}}(V)/p^r. \quad (28)$$

If a finite range $0 \leq n \leq N$ is fixed and $|k| > \max_{n \leq N} M_n$, the same is true with ordinary Milnor K -theory. In particular

$$q_1 : \ker(K_n^{\mathbf{M}}(T)/p \xrightarrow{q_0} A_n) \xrightarrow{\sim} J_n. \quad (29)$$

Proof. Lüdgers–Morrow prove for every local p -henselian ring R that the syntomic symbol is an isomorphism

$$\widehat{K}_n^{\mathbf{M}}(R)/p^r \xrightarrow{\sim} H^n(\mathbf{Z}/p^r(n)_{\text{syn}}(R)) \quad (30)$$

[12, Theorem 3.1]. The rings V , k , and T are henselian; for T , a simple residue root lifts in the two V -coordinates and the lifts have the same residue. Moreover $\mathbf{Z}/p^r(n)_{\text{syn}}(k) \in D^{\geq n}$ by the logarithmic de Rham–Witt description in characteristic p [6, Proposition 8.4 and Corollary 8.21]. Taking H^n of theorem 6.2 therefore gives exactly the fiber product in (28).

For a fixed finite range, (20) converts all four improved groups into ordinary ones. Finally, the quotient of the Milnor presentation of V by the ideal generated by P is the Milnor presentation of k : units lift and every residue Steinberg relation lifts. Hence $J_n = \ker(A_n \rightarrow K_n^{\mathbf{M}}(k)/p)$, and (29) follows. $\square$

6.3 A universal degree-three class

The only genuinely new operation is constructed in degree three. The higher operations in the next section will be forced by it.

Lemma 6.4 (Universal three-variable class). *Let R be a henselian local $\mathbf{Z}_p$ -algebra and let $x, y, z \in R$ with $1 + xyz \in R^\times$. There is a natural class*

$$\mathcal{D}_R(x; y, z) \in \widehat{K}_3^M(R)/p$$

which equals $\{1 + xyz, y, z\}$ when y, z are units. It is functorial in R and satisfies

$$\mathcal{D}(x; y, z) = -\mathcal{D}(x; z, y), \quad (31)$$

$$\mathcal{D}(x; y, z_1 z_2) = \mathcal{D}(x z_2; y, z_1) + \mathcal{D}(x z_1; y, z_2), \quad (32)$$

$$\mathcal{D}(x; y, z h^p) = \mathcal{D}(x h^p; y, z), \quad (33)$$

$$\mathcal{D}(x; y, u) = \langle x u, y \rangle \{u\} \quad (u \in R^\times), \quad (34)$$

$$\mathcal{D}(x; y, y) = \mathcal{D}(-x y; y, -1). \quad (35)$$

In (34) we use the identification $K_2^M \simeq K_2$ whenever the residue field is large enough. All formulas remain valid for $p = 2$.

Proof. On

$$X = \operatorname{Spec} \mathbf{Z}_p[X, Y, Z, (1 + XYZ)^{-1}]$$

the rational Milnor class $\{1 + XYZ, Y, Z\}$ has zero residue at every codimension-one point. The only possible nonunit divisors are $Y = 0$ and $Z = 0$, where the first entry specializes to 1; the vertical divisor causes no additional residue. The Gersten theorem for improved Milnor K -theory on smooth schemes over a complete DVR therefore gives a unique Nisnevich section extending this rational class; see [12, Theorem 4.1 and Corollary 4.4]. Pulling its henselian stalk to R defines $\mathcal{D}_R$.

We record why the identities can be specialized without using the endpoint theorem that will only be proved later. Fix one of the desired polynomial identities among the $\mathcal{D}$ -classes. Choose a smooth polynomial localization over $\mathbf{Z}_p$ in all of the displayed variables, inverting exactly the finitely many expressions that must be units. At its generic point the $\mathcal{D}$ -classes are ordinary Milnor symbols, so the identity follows from the standard multilinear and Steinberg relations. The Gersten injection of [12, Corollary 4.4] then shows that the equality already holds as a global improved-Milnor section on this smooth source. Only after this step do we specialize to the henselian local ring R . In particular we never use an injection of $\widehat{K}_3^M(R)/p$ into the Milnor group of its fraction field.

For (31) this is skew symmetry in the last two entries. For (32) it is expansion of the last Milnor entry. Before reduction modulo p , the same calculation gives

$$\mathcal{D}(x; y, z h^p) = \mathcal{D}(x h^p; y, z) + p \mathcal{D}(x z h^{p-1}; y, h),$$

which proves (33). Formula (35) comes from the integral identity $\{y, y\} = \{y, -1\}$; it does not use antisymmetrization or division by 2.

Formula (34) deserves a separate check because its right hand side involves a Dennis–Stein K_2 class. In the applications below the residue field is chosen large enough that ordinary Milnor K_2 agrees with Quillen K_2 . To prove the universal identity, take a smooth local polynomial source

with residue field in that range; if necessary one may adjoin finitely many auxiliary variables and specialize them to distinct residue classes. On this source the Dennis–Stein unit formula identifies

$$\langle xu, y \rangle = \{1 + xyu, y\} \in K_2,$$

so after multiplying by $\{u\}$ both sides of (34) agree at the generic point. The same smooth-source Gersten injection makes them equal before specialization. This is why (34) is available on V without importing any unproved relative presentation or any endpoint injectivity. All of these arguments are integral before the final reduction modulo p , and therefore remain valid for $p = 2$. $\square$

6.4 Relative degree-two and degree-three operations

We first identify the relative K_2 group with the principal-unit ideal in A_2 .

Lemma 6.5. *Assume $|k| > 5$ and that ordinary and improved Milnor K -theory agree in degrees at most 3. Then the natural map identifies*

$$K_2(V, \mathfrak{m})/p \xrightarrow{\sim} J_2 \subset A_2.$$

Proof. The coefficient sequence for relative K -theory shows that it is enough to know that

$$K_3(V; \mathbf{F}_p) \longrightarrow K_3(k; \mathbf{F}_p)$$

is surjective. Geisser–Levine identify $K_3(k; \mathbf{F}_p)$ with the image of Milnor symbols [9, Theorem 8.1]; lifting the entries of such symbols to units of V gives the required surjectivity. The relative coefficient sequence therefore makes

$$K_2(V, \mathfrak{m}; \mathbf{F}_p) \longrightarrow K_2(V; \mathbf{F}_p)$$

injective. The universal-coefficient sequences contain natural injections of the integral groups modulo p into the corresponding coefficient groups, so $K_2(V, \mathfrak{m})/p \rightarrow K_2(V)/p$ is injective as well.

By Matsumoto’s theorem for the field k , $K_2(k) = K_2^M(k)$, and every residue symbol lifts to V ; hence $K_2(V) \rightarrow K_2(k)$ is surjective. Reducing the relative exact sequence modulo p therefore identifies the image of the relative group with the kernel of $K_2(V)/p \rightarrow K_2(k)/p$. Using (24) for V , that kernel is the kernel of $K_2^M(V)/p \rightarrow K_2^M(k)/p$, namely $PA_1 = J_2$. $\square$

For $g \in G$ and a principal unit $w = \{1 + a\} \in P$, choose a representative $\rho \in K^\times$ of g with $0 < v(\rho) < v(a)$ and set

$$d_g(w) = \left\langle \frac{a}{\rho}, \rho \right\rangle \in J_2. \tag{36}$$

The density lemma guarantees such ρ .

Lemma 6.6. *Formula (36) induces an $\mathbf{F}_p$ -bilinear map*

$$d : G \times P \longrightarrow J_2.$$

If $u \in V^\times$ represents a class of $A_1 \subset G$, then

$$d_{\{u\}}(w) = w\{u\}. \tag{37}$$

Proof. Relations (21)–(23) give, for a common sufficiently small ρ ,

$$\left\langle \frac{a}{\rho}, \rho \right\rangle + \left\langle \frac{b}{\rho}, \rho \right\rangle = \left\langle \frac{a+b+ab}{\rho}, \rho \right\rangle,$$

so the construction is additive in the principal unit. If $\rho' = \rho h^p$ represents the same element of G , order the two values so that h is integral. Iterating (23) gives

$$\left\langle \frac{a}{\rho'}, \rho' \right\rangle = \left\langle \frac{a}{\rho}, \rho \right\rangle + p \left\langle \frac{a}{h}, h \right\rangle,$$

so the value is unchanged modulo p . The same relation, applied to a product of two small representatives, gives additivity in g . Finally take $\rho = uh^p$ with small positive value and use the unit-entry Dennis–Stein formula to obtain (37). $\square$

We now construct the degree-three operation. If $a \in \mathfrak{m}$ define

$$f_g(a, b) = \mathcal{D}(a/\rho; b, \rho), \quad (38)$$

and if $b \in \mathfrak{m}$ define

$$f_g(a, b) = -\mathcal{D}(-b/\rho; -a, \rho), \quad (39)$$

where ρ is a sufficiently small positive representative of g .

Lemma 6.7. *The two formulas agree when both entries are in $\mathfrak{m}$, are independent of the chosen ρ , and are additive in g .*

Proof. Write $a = \rho A$, $b = \rho B$. The difference between (38) and (39) is, on the universal smooth source, the Milnor class

$$\{1 + \rho^2 AB, -\rho^2 AB, \rho\} = 0;$$

theorem 6.4 extends this identity to the specialization. Independence under $\rho \mapsto \rho h^p$ is (33). Additivity follows from (32) after choosing representatives ρ, σ with product sufficiently small. $\square$

Proposition 6.8 (Degree-three relative operation). *For every $g \in G$ there is an $\mathbf{F}_p$ -linear map*

$$\Theta_g : J_2 \longrightarrow A_3,$$

additive in g , such that for $w \in P$, $a \in A_1$, and $g, h \in G$,

$$\Theta_g(wa) = -d_g(w)a, \quad (40)$$

$$\Theta_h d_g(w) = -\Theta_g d_h(w), \quad (41)$$

$$\Theta_g d_g(w) = d_g(w)e. \quad (42)$$

At $p = 2$, the e -term in (42) is essential.

Proof. We first construct a homomorphism on the absolute group $K_2(T)$. Write a Dennis–Stein symbol as

$$r = (x, x+a), \quad t = (y, y+b), \quad a, b \in \mathfrak{m},$$

and put $U = 1 + xy$ and $H = 1 + x(y+b)$. Both are units whenever the endpoint Dennis–Stein symbols are defined. Set

$$\Phi_g \langle r, t \rangle = f_g \left(x, \frac{b}{U} \right) + f_g \left(\frac{a}{H}, y+b \right). \quad (43)$$

For a finite relation choose one representative ρ small enough to divide all coordinate differences. On a smooth universal localization, (43) has generic value

$$(\langle x + a, y + b \rangle - \langle x, y \rangle)\{\rho\}. \quad (44)$$

Indeed, the usual first- and second-variable Dennis–Stein expansions give

$$\langle x + a, y + b \rangle - \langle x, y \rangle = \left\langle x, \frac{b}{1 + xy} \right\rangle + \left\langle \frac{a}{1 + x(y + b)}, y + b \right\rangle,$$

and [theorem 6.4](#) identifies multiplication by $\{\rho\}$ with the two terms in (43).

We now verify the full Dennis–Stein presentation, including the intermediate denominators. For a finite relation choose one common positive representative ρ small enough to divide every coordinate difference that appears. Write

$$r = (x, x + \rho A), \quad t = (y, y + \rho B), \quad u = (z, z + \rho C).$$

The formula (43) is represented on the smooth polynomial localization in x, y, z, A, B, C, ρ obtained by inverting the endpoint denominators and the intermediate terms occurring below. Every such intermediate denominator has the same residue as the corresponding endpoint denominator; hence it specializes to a unit of V . The generic value is (44).

For (21), the generic image of

$$\langle r, t \rangle + \langle -t, -r \rangle$$

is

$$(\langle x + \rho A, y + \rho B \rangle + \langle -y - \rho B, -x - \rho A \rangle - \langle x, y \rangle - \langle -y, -x \rangle)\{\rho\},$$

which is zero by the same DS1 relation in the two coordinates.

For (22), put $w = t + u + rtu$. Its first coordinate is

$$w_0 = y + z + xyz,$$

and its second coordinate is $w_0 + \rho W$, where

$$W = B + C + Ayz + xBz + xyC + \rho(ABz + AyC + xBC) + \rho^2 ABC.$$

The generic image of

$$\langle r, t \rangle + \langle r, u \rangle - \langle r, w \rangle$$

is exactly the DS2 relation for $(x + \rho A, y + \rho B, z + \rho C)$ minus the DS2 relation for (x, y, z) , multiplied by $\{\rho\}$. Hence it vanishes.

For (23), the coordinate differences of tu , rt , and ru are respectively

$$\rho(yC + zB + \rho BC), \quad \rho(xB + yA + \rho AB), \quad \rho(xC + zA + \rho AC).$$

Therefore the generic image of

$$\langle r, tu \rangle - \langle rt, u \rangle - \langle ru, t \rangle$$

is the difference of the two DS3 relations in the second and first coordinates, again multiplied by $\{\rho\}$, and is zero.

Each of the three generic equalities is an equality of the universal $\mathcal{D}$ -classes on the chosen smooth localization. By [theorem 6.4](#), Gersten injectivity makes the equality hold before specialization, including specializations with zero entries or nonunits. Independence of the functions f_g from the

representative allows the common ρ to be chosen for any finite relation. Thus (43) respects the complete absolute Dennis–Stein presentation and defines a homomorphism

$$\Phi_g : K_2(T) \rightarrow A_3.$$

When $a = b = 0$ the right hand side of (43) is zero, so Φ_g vanishes on the diagonal image $s_* K_2(V)$.

Restrict Φ_g to the kernel of q_0 . By theorems 6.1 and 6.5, this kernel modulo p identifies canonically with J_2 through q_1 ; the resulting map is Θ_g . For $w = \{1 + a\}$ and a unit u , the two evident lifts in the double reduce the formulas to

$$\Theta_g(w\{u\}) = \mathcal{D}(a/(u\rho); u, \rho) = -\langle a/\rho, \rho \rangle \{u\},$$

using (31) and (34). This proves (40).

Choose representatives ρ, σ of g, h with $v(\rho) + v(\sigma) < v(a)$. The definitions give

$$\Theta_h d_g(w) = \mathcal{D}(a/(\rho\sigma); \rho, \sigma),$$

and (31) proves (41). If $g = h$, choose $2v(\rho) < v(a)$; then (35) and (34) give

$$\Theta_g d_g(w) = \mathcal{D}(a/\rho^2; \rho, \rho) = \mathcal{D}(-a/\rho; \rho, -1) = d_g(w)e.$$

No division by 2 occurs. □

7 All-degree Milnor operations and the endpoint

The degree-three identities in theorem 6.8 contain exactly the coherence needed to extend the operation to every degree. We separate the formal graded-algebra argument from the valuation-theoretic application.

7.1 A graph normal form

Let $A = \bigoplus_{m \geq 0} A_m$ be a connected graded-commutative $\mathbf{F}_p$ -algebra generated by A_1 . Fix $e \in A_1$ such that

$$u^2 = ue \quad (u \in A_1), \tag{45}$$

with $e = 0$ when p is odd. Let $P \subset A_1$, $J = AP$, and let $G \supset A_1$ be an $\mathbf{F}_p$ -vector space. Choose a complement with ordered basis $(t_i)_{i \in I}$ and let C be the graded-commutative A -algebra generated by the t_i subject to

$$t_i^2 = t_i e. \tag{46}$$

For a finite ordered set $S = \{i_1 < \dots < i_r\}$ write $t_S = t_{i_1} \dots t_{i_r}$.

Lemma 7.1. *As a left or right A -module,*

$$C = \bigoplus_{S \subset I, |S| < \infty} At_S.$$

Consequently $JC = \bigoplus_S Jt_S$ and $A \cap JC = J$.

Proof. For odd p this is the usual squarefree basis of an exterior algebra. For $p = 2$, adjoining one generator amounts to quotienting by the monic polynomial $T^2 - eT$, so the result is free with basis $1, T$. Adjoin finitely many variables successively and pass to the filtered union. No condition on e being a nonzerodivisor is needed. □

Lemma 7.2 (Graph normal form). *Suppose that for every i there are degree-one linear maps*

$$R_i : J_m \rightarrow J_{m+1} \quad (m \geq 1)$$

such that, for homogeneous $q \in J$, $a \in A$,

$$R_i(qa) = (-1)^{\deg a} R_i(q)a, \quad (47)$$

$$R_j R_i(q) = -R_i R_j(q), \quad (48)$$

$$R_i^2(q) = R_i(q)e. \quad (49)$$

Put $d_i = R_i|_P$ and

$$B = C / (wt_i - d_i(w) : w \in P, i \in I).$$

Then $A \rightarrow B$ is injective. More precisely, for any graded vector-space splitting $A = J \oplus H$,

$$B \cong A \oplus \bigoplus_{\emptyset \neq S \subset I} Ht_S \quad (50)$$

as graded vector spaces.

Proof. Give J its usual right A -module structure and prescribe right multiplication by t_i to be R_i . Equations (47)–(49) are exactly the relations required for this action to descend from the free associative extension to C . Thus there is a right C -linear evaluation map

$$\epsilon : JC = J \otimes_A C \longrightarrow J, \quad q \otimes c \longmapsto q \cdot c,$$

which restricts to the identity on J .

We claim that $\ker \epsilon$ is generated by $qt_i - R_i(q)$, $q \in J$. Expand an element of JC in the squarefree basis of [theorem 7.1](#). For a term $qt_{i_1} \cdots t_{i_r}$, successively subtract

$$(qt_{i_1} - R_{i_1}(q))t_{i_2} \cdots t_{i_r},$$

then the analogous expression with $R_{i_1}(q)$ and t_{i_2} , and so on. This telescopes to

$$qt_S - R_{i_r} \cdots R_{i_1}(q) = qt_S - \epsilon(qt_S).$$

Hence every x differs from $\epsilon(x)$ by the stated generators, and the reverse containment is clear.

It is enough to take $q \in P$ in these generators. Indeed (47) and graded commutativity imply $R_i(aq) = aR_i(q)$; writing q as a sum of A -multiples of elements of P reduces the general generator to the defining graph relations. Thus the quotient ideal is exactly $\ker \epsilon$.

If $a \in A$ dies in B , then $a \in A \cap \ker \epsilon \subset A \cap JC = J$, while $\epsilon(a) = a$; hence $a = 0$. Splitting the coefficient of every nonconstant t_S into its J - and H -parts and evaluating the J -part gives (50); the same argument proves linear independence. $\square$

For an application in degree n , only operations through degree $n - 1$ are needed. Equivalently, one may truncate A above degree n and set the operations to zero in the top degree; all omitted identities then land beyond the truncation. We will use this finite-range form without further comment.

7.2 Extending the degree-three operation

The following elementary lemma is the mechanism that turns d_g and Θ_g into operations on all Milnor degrees.

Lemma 7.3 (Twisted derivation extension). *Let $M = T(E)/(Q)$ be a connected quadratic graded algebra with E in degree one. Let $f_0, f_1 : M \rightarrow A$ be graded algebra maps such that*

$$w_x = f_1(x) - f_0(x) \in P \quad (x \in E).$$

Suppose maps $d : P \rightarrow J_2$ and $\Theta : J_2 \rightarrow A_3$ satisfy

$$\Theta(wa) = -d(w)a \quad (w \in P, a \in A_1). \quad (51)$$

Put $\delta(x) = d(w_x)$. Then

$$D(x_1 \cdots x_m) = \sum_{i=1}^m (-1)^{m-i} f_1(x_1) \cdots f_1(x_{i-1}) \delta(x_i) f_0(x_{i+1}) \cdots f_0(x_m) \quad (52)$$

defines maps $D : M_m \rightarrow J_{m+1}$. They satisfy

$$D(uv) = f_1(u)D(v) + (-1)^{\deg v} D(u)f_0(v). \quad (53)$$

If a section $s : A \rightarrow M$ satisfies $f_0s = f_1s = \text{id}$, then

$$D(s(a)) = 0, \quad D(zs(a)) = (-1)^{\deg a} D(z)a. \quad (54)$$

Proof. Formula (52) first defines a multilinear map on the tensor algebra, and separating the differentiated slot between two words gives (53). For $x, y \in E$,

$$f_1(x)f_1(y) - f_0(x)f_0(y) = w_x f_0(y) + f_1(x)w_y \in J_2.$$

Using (51) and graded commutativity,

$$D(xy) = \Theta(f_1(x)f_1(y) - f_0(x)f_0(y)).$$

If $r \in Q$, both $f_0(r)$ and $f_1(r)$ vanish, hence $D(r) = 0$. The Leibniz rule then gives $D(urv) = 0$ for arbitrary words u, v , so D kills the entire two-sided relation ideal and descends to M . Each term contains one $\delta(x_i) \in J_2$, hence lands in J . Finally $w_{s(a)} = 0$ on degree-one diagonal elements; generation in degree one and (53) give (54). $\square$

Apply this with

$$M = K_*^{\text{M}}(T)/p, \quad f_i = q_i, \quad s : K_*^{\text{M}}(V)/p \rightarrow K_*^{\text{M}}(T)/p,$$

and with $(d, \Theta) = (d_g, \Theta_g)$. Milnor K -theory modulo p has the required quadratic tensor-algebra presentation by the Steinberg relations. For a unit pair $t_i = (u_i, u'_i) \in T^\times$, the resulting operation is

$$\Phi_g^{(m)}\{t_1, \dots, t_m\} = \sum_{i=1}^m (-1)^{m-i} \{u'_1, \dots, u'_{i-1}\} d_g\{u'_i/u_i\} \{u_{i+1}, \dots, u_m\}. \quad (55)$$

It vanishes on diagonal classes and satisfies the module rule (54).

We now explain carefully why these absolute operations descend to operations on the actual ideal J , rather than merely on a presentation of it. Put

$$L_m = \ker(q_0 : K_m^M(T)/p \rightarrow A_m).$$

The independently proved Milnor excision isomorphism (29) says that

$$q_1 : L_m \xrightarrow{\sim} J_m \quad (56)$$

for every degree in the chosen finite range. Thus every $\alpha \in J_m$ has a unique relative lift $\tilde{\alpha} \in L_m$. Restrict $\Phi_g^{(m)}$ to L_m and define

$$R_g(\alpha) = \Phi_g^{(m)}(\tilde{\alpha}) \in J_{m+1}. \quad (57)$$

The target lies in J_{m+1} because every summand of (55) contains the factor $d_g(w_i) \in J_2$. This construction uses the genuine double-excision isomorphism (56); it does not infer the existence of R_g from generators and syzygies of the ideal J . In particular all higher relations in J_m are automatically respected.

Proposition 7.4 (All-degree operations). *In every fixed finite degree range, the maps (57) are bilinear in g and satisfy*

$$R_g(w) = d_g(w) \quad (w \in P), \quad (58)$$

$$R_g(\alpha a) = (-1)^{\deg a} R_g(\alpha) a, \quad (59)$$

$$R_u(\alpha) = \alpha u \quad (u \in A_1), \quad (60)$$

$$R_h R_g(\alpha) = -R_g R_h(\alpha), \quad (61)$$

$$R_g^2(\alpha) = R_g(\alpha) e. \quad (62)$$

The identities include $p = 2$.

Proof. Let $w = \{1 + a\} \in P$. Its canonical relative lift is the degree-one class of the unit pair $(1, 1 + a) \in T^\times$. Formula (55) in degree one therefore gives

$$R_g(w) = d_g(w),$$

which proves (58). If $\tilde{\alpha} \in L_m$ is the unique lift of $\alpha \in J_m$, then $\tilde{\alpha} s_*(a)$ is the unique relative lift of αa : its first coordinate is zero and its second coordinate is αa . Equation (54) consequently gives

$$R_g(\alpha a) = (-1)^{\deg a} R_g(\alpha) a,$$

so the module rule is an identity on the actual ideal, not merely on chosen generators.

Every $\alpha \in J_m$ is a sum of terms wa with $w \in P$ and $a \in A_{m-1}$. If $u \in A_1 \subset G$, then (37) and the module rule give

$$R_u(wa) = (-1)^{m-1} d_u(w) a = (-1)^{m-1} wua = wau,$$

where the last equality is graded commutativity in A . This proves (60).

We next identify the degree-two operation. For $wa \in J_2$ with $w \in P$, $a \in A_1$, the module rule gives

$$R_g(wa) = -d_g(w) a.$$

This is exactly the defining formula (40) for Θ_g . Since such products span J_2 , we have $R_g|_{J_2} = \Theta_g$. Hence the degree-three identities already proved give

$$R_h d_g(w) = -R_g d_h(w), \quad R_g d_g(w) = d_g(w)e. \quad (63)$$

No higher-degree injectivity statement is used here; these are the identities obtained from the universal $\mathcal{D}$ -calculation in the preceding section.

Finally take a homogeneous $a \in A$. Applying the module rule twice to a generator $wa \in J$ gives

$$R_h R_g(wa) = R_h d_g(w) a, \quad R_g R_h(wa) = R_g d_h(w) a,$$

with the same total Koszul sign on the two expressions; (63) proves anticommutation. Likewise

$$R_g^2(wa) = R_g d_g(w) a = d_g(w)e a = R_g(wa)e.$$

Since the elements wa span J , this proves (61) and (62) in every degree under consideration. At $p = 2$ the signs become plus signs, but the relation $R_g^2 = R_g(-)e$ remains unchanged; the $e = \{-1\}$ term must not be discarded. $\square$

7.3 Endpoint injectivity

Theorem 7.5 (Milnor endpoint). *Let V be any complete rank-one valuation ring of mixed characteristic $(0, p)$. For every $n \geq 1$, the natural map*

$$\widehat{K}_n^M(V)/p \longrightarrow K_n^M(\text{Frac } V)/p \quad (64)$$

is injective. Under the syntomic symbol and the norm-residue isomorphism, this is the injectivity of

$$H^n S_n(V) \longrightarrow H^n E_n(V). \quad (65)$$

Proof. Assume first that V is nondiscrete and that k is sufficiently large. Fix $N \geq \max\{3, n\}$ so that the ordinary/improved comparison holds through degree N . Set $A = K_*^M(V)/p$ in this range. The inclusion $A_1 \hookrightarrow G$ is injective: if a unit is a p th power in K , its p th root has value zero and is again a unit. Choose a vector-space complement $G = A_1 \oplus \bigoplus_i \mathbf{F}_p t_i$.

The relations (59), (61), and (62) satisfy the hypotheses of theorem 7.2; moreover $u^2 = ue$ in A_1 by (25). We therefore obtain a graded algebra B containing A injectively and generated over A by the new degree-one classes t_i , with graph relations

$$wt_i = d_{t_i}(w), \quad t_i^2 = t_i e.$$

Equivalently, every $g \in G$ acts as a degree-one generator and satisfies the corresponding relations.

It remains to verify that every field Steinberg relation $[x][1-x] = 0$ holds in B_2 . If $v(x) > 0$, choose a representative ρ of $[x] \in G$ with $0 < v(\rho) < v(x)$ and write $x = \rho h^p$. Since $1-x$ is a principal unit, the graph relation and Dennis–Stein identities give

$$[x][1-x] = d_{[x]}(1-x) = \langle -h^p, \rho \rangle = -\langle -\rho, h^p \rangle = -p \langle -\rho h^{p-1}, h \rangle = 0.$$

If $v(1-x) > 0$, interchange x and $1-x$. If both are units, the Steinberg relation already holds in A_2 . Finally suppose $v(x) < 0$ and put $z = x^{-1}$. In G ,

$$[1-x] = e + [x] + [1-z],$$

while $[x][1 - z] = -[z][1 - z] = 0$ by the preceding unit case. Hence

$$[x][1 - x] = [x]e + [x]^2 = 2[x]e = 0 :$$

for odd p one has $e = 0$, while for $p = 2$ the scalar 2 vanishes. Thus the Milnor presentation of the field gives a graded map

$$K_*^M(K)/p \longrightarrow B$$

whose restriction to A_n is the already injective map $A_n \hookrightarrow B_n$. This proves (64) for large residue field.

If k is finite, only finitely many residue-size bounds are relevant, namely those needed for the ordinary/improved comparison through degree N . Choose a finite residue extension k'/k of degree d prime to p whose cardinality exceeds all of them, and lift it by henselianity to a finite etale local valuation extension V'/V of the same degree. Its value group is unchanged, so the nondiscrete argument applies to V' . If a class $x \in H^n S_n(V)$ dies generically, its restriction to V' dies generically and hence is zero by the large-residue endpoint. The finite etale syntomic norm of [12, Lemma 3.3 and Corollary 3.4] satisfies

$$\mathrm{Nm}_{V'/V} \circ \mathrm{res}_{V'/V} = d$$

on syntomic cohomology. Since d is invertible modulo p , restriction is injective, and therefore $x = 0$. This proves the endpoint without imposing a large-residue hypothesis on V .

If V is discrete, it is a complete DVR and the endpoint is part of the regular F -smooth comparison theorem [4, Theorem 4.15 and Proposition 5.2].

Finally, the syntomic symbol (30) identifies the source with $H^n S_n(V)$, and the norm-residue theorem identifies $K_*^M(K)/p$ with $H_{\mathrm{et}}^n(K, \mu_p^{\otimes n})$ [18, Theorem 6.16]. The maps agree with the generic Galois symbol, proving (65). $\square$

8 Frobenius base change and twist periodicity

The remaining obstruction is the family of completion defects $\mathcal{E}_k$. We now show that, after one conservative Frobenius-type base change, all twists become equivalent. This is the only place where the geometry of the Hodge–Tate stack is used.

Put

$$A = V/p, \quad \psi : V \longrightarrow A, \quad v \longmapsto \bar{v}^p,$$

and write $\psi^*(-) = - \otimes_{V, \psi}^{\mathbf{L}} A$. For this section V is nondiscrete, complete and rank one.

8.1 Finite-Tor base change

Lemma 8.1 (Eventual-stabilization base change). *Let B have Tor amplitude $[-d, 0]$ over a ring R . Let (X_i) be an inverse system such that, for every i , the fibers $\mathrm{fib}(X_j \rightarrow X_i)$ for $j \geq i$ lie in $\mathcal{D}^{\geq c_i}$ with $c_i \rightarrow \infty$. Then*

$$B \otimes_R^{\mathbf{L}} \mathrm{Rlim}_i X_i \xrightarrow{\sim} \mathrm{Rlim}_i (B \otimes_R^{\mathbf{L}} X_i).$$

The same applies to the partial-totalization tower of a cosimplicial object whose terms have a uniform lower cohomological bound.

Proof. Let $\widehat{X} = \operatorname{Rlim}_i X_i$. Taking the inverse limit over $j \geq i$ in the fiber triangles gives $\operatorname{fib}(\widehat{X} \rightarrow X_i) \in D^{\geq c_i}$. Tensoring with B lowers connectivity by at most d . The same estimate holds for the limit of the base-changed tower. Hence the canonical map is an isomorphism in each fixed cohomological degree for $i \gg 0$. For totalizations, normalized cosimplicial degree r contributes with a shift by $-r$, so the fibers of consecutive partial totalizations become arbitrarily connective; apply the first assertion. $\square$

Lemma 8.2 (Conservativity and one-degree detection). *The functor $F_A^* : D(A) \rightarrow D(A)$ is conservative. Consequently $\psi^* : D(V) \rightarrow D(A)$ is conservative on derived p -complete complexes. The V -module A via ψ has Tor amplitude $[-1, 0]$.*

Moreover, if a V -module M is annihilated by a power of p and

$$\operatorname{Tor}_1^V(M, A; \psi) = 0,$$

then $M = 0$.

Proof. A valuation ring has weak dimension at most one, so every V -module has flat dimension at most one; this gives the Tor-amplitude assertion.

Extend the valuation to $K(p^{1/p})$ and let V' be the resulting valuation ring. It is faithfully flat over V . Choose $\pi \in V'$ with $\pi^p = p$ and put $A' = V'/p$. Frobenius on A' factors as

$$A' \longrightarrow V'/\pi \longrightarrow A'.$$

The first arrow has nilpotent kernel. The second, $x \mapsto x^p$, is faithfully flat: V'/π is a quotient of a valuation ring, hence its finitely generated ideals are principal, and the annihilator criterion for flatness applies exactly as in [13, Proposition A.4 and Corollary A.5]; see also [10]. Derived base change by a nilpotent quotient is conservative, so $F_{A'}^*$ is conservative. Faithfully flat descent along $A \rightarrow A'$ then proves conservativity of F_A^* .

For a derived p -complete N ,

$$\psi^* N \simeq F_A^*(N \otimes_V^{\mathbf{L}} A).$$

Thus $\psi^* N = 0$ implies $N/p = 0$, and derived p -completeness gives $N = 0$.

For the last statement, base change to V' and factor ψ' through $B = V'/\pi$ followed by the faithfully flat map $B \rightarrow A'$. The two-term resolution of B over V' gives

$$\operatorname{Tor}_1^{V'}(M \otimes_V V', A'; \psi') \simeq (M \otimes_V V')[\pi] \otimes_B A'.$$

Its vanishing implies that multiplication by π is injective on $M \otimes_V V'$. But $p^N M = 0$ for some N , hence π^{p^N} acts by zero. An injective nilpotent endomorphism forces the module to vanish; faithful flatness then gives $M = 0$. $\square$

8.2 The actual twisted Nygaard filtrations

We use the multiplicative induced twisted Nygaard filtration of Bhatt–Mathew [4, Constructions 3.1 and 3.3, Remark 3.4]. It is important here to use the actual filtered complexes, not merely their associated graded objects. For every $k \in \mathbf{Z}$,

$$\mathcal{F}^0 H_k = H_k, \quad \mathcal{F}^r H_k \otimes \mathcal{F}^s H_\ell \longrightarrow \mathcal{F}^{r+s} H_{k+\ell}. \quad (66)$$

The all-integer twist and multiplicativity statements follow from the filtered Breuil–Kisin twists in [2, Section 5.5]. On QRSP charts this is simply the filtration induced by the images of the Nygaard ideals in Hodge–Tate cohomology, and the finite quotients are the B_q^k of section 4.

Lemma 8.3 (Completion commutes with ψ^*). *For every k ,*

$$\psi^* \widehat{H}_k \xrightarrow{\sim} \operatorname{Rlim}_q \psi^* B_q^k.$$

Consequently $\psi^ \mathcal{E}_k$ is the completion defect of the base-changed filtered object $\psi^* H_k$.*

Proof. By [theorem 3.6](#), the twisted Bott graded fibers become increasingly connective. More precisely, the induced twisted filtration satisfies

$$\operatorname{gr}_{\mathcal{F}}^j H_k \in D^{\geq j-p+1} \quad (j \geq p),$$

uniformly in k ; this is exactly the cohomological form of weak F -smoothness used in [\[4, Remark 4.5\]](#). Hence for $q' \geq q \geq p$,

$$\operatorname{fib}(B_{q'}^k \rightarrow B_q^k) \in D^{\geq q-p+1}.$$

The V -module A via ψ has Tor amplitude $[-1, 0]$ by [theorem 8.2](#). Apply [theorem 8.1](#); exactness of ψ^* then identifies the cofibers as claimed. $\square$

If a twist-one class becomes a unit after ψ -base change, [\(66\)](#) makes it a filtered unit automatically.

Lemma 8.4 (Filtered unit). *Suppose $\eta \in H^0(\psi^* H_1)$ and $\eta^{-1} \in H^0(\psi^* H_{-1})$ multiply to 1. Multiplication by η gives inverse equivalences of filtered objects*

$$\psi^*(\mathcal{F}^\bullet H_k) \xrightarrow{\sim} \psi^*(\mathcal{F}^\bullet H_{k+1})$$

for all k , and hence equivalences of their completion defects.

Proof. Both classes live in filtration zero because $\mathcal{F}^0 H_{\pm 1} = H_{\pm 1}$. Multiplicativity gives maps on every filtration step. Their products are the identity in $H^0(\mathcal{F}^0 H_0)$, so the two filtered multipliers are inverse in the filtered derived category. Finite quotients, inverse limits, and cofibers preserve the equivalence; [theorem 8.3](#) identifies the resulting defects with $\psi^* \mathcal{E}_k$. $\square$

8.3 A geometric trivialization of the Breuil–Kisin line

We now construct the required unit. The quotient presentation of the Hodge–Tate prismatization for a regular local prism in [\[3, Construction 9.4 and Proposition 9.5\]](#) is the geometric input. We use only the following consequence: for a bounded prism $(P, (E))$ presenting a complete regular local ring, the Hodge–Tate Cartier–Witt stack is a quotient of a $\mathbf{G}_a^\sharp$ -torsor by the divided-power tangent group semidirect $\mathbf{G}_m^\sharp$, and the $\mathbf{G}_m^\sharp$ -character on the Breuil–Kisin line is the zeroth Witt coordinate; cf. also the Hodge–Tate description of the twist in [\[2, Section 4.4\]](#).

Lemma 8.5. *There exist $a, b \in \mathfrak{m}_V$ such that*

$$a^p b = p, \quad a^{p^2} \in pV.$$

For

$$P = \mathbf{Z}_p[[T, B]], \quad E = p - T^p B, \quad R = P/(E),$$

with $\delta(T) = \delta(B) = 0$, the pair $(P, (E))$ is a bounded prism, R is a p -torsion-free complete regular local ring of dimension two, and $T \mapsto a, B \mapsto b$ defines a continuous map $R \rightarrow V$.

Proof. Normalize $v(p) = 1$. Density of the nondiscrete value group lets us choose

$$\frac{1}{p^2} \leq v(a) < \frac{1}{p}, \quad b = p/a^p.$$

Then $a, b \in \mathfrak{m}$, $a^p b = p$, and $a^{p^2} \in pV$.

The ring P is regular local of dimension three and $E \in (p, T, B) \setminus (p, T, B)^2$, so R is regular local of dimension two. Modulo p , $E = -T^p B$ is a nonzerodivisor in $\mathbf{F}_p[[T, B]]$, which implies that R is p -torsion-free. Finally $\delta(E)$ has unit constant term $\delta(p) = 1 - p^{p-1}$; hence $p \in (E, \varphi(E))$. Completeness and boundedness are immediate, so $(P, (E))$ is a bounded prism. Convergence of the evaluation map follows from $v(a), v(b) > 0$. $\square$

Lemma 8.6 (Hodge–Tate global sections after ψ -base change). *Let $X = \mathrm{Spf} V$, let X^{HT} be its absolute Hodge–Tate stack, and put*

$$Z = \mathrm{Spec} A \times_{X, \psi} X^{\mathrm{HT}}$$

with derived fiber product. For every $k \in \mathbf{Z}$, the global sections of the pulled k th Breuil–Kisin line identify multiplicatively with

$$R\Gamma(Z, \mathcal{O}\{k\}) \simeq \psi^*(\overline{\Delta}_V\{k\}). \quad (67)$$

Consequently a trivialization of the line and of its dual on Z gives inverse classes after the canonical derived reduction

$$\psi^*(\overline{\Delta}_V\{\pm 1\}) \longrightarrow \psi^*H_{\pm 1}.$$

Proof. Choose the perfectoid valuation cover $V \rightarrow W$ from [theorem 2.3](#), and let

$$R^a = (W^{\otimes_V(a+1)})_p^\wedge$$

be its Čech terms. They are p -torsion-free QRSP. By the prismatization comparison and flat-local descent [[3](#), Lemma 6.3, Theorem 7.17, Corollary 7.18 and Corollary 8.17], the Hodge–Tate stack X^{HT} is covered by the derived formal affines with rings $\overline{\Delta}_{R^a}$, with the displayed cosimplicial object as the full Čech nerve. The k th Breuil–Kisin line has a discrete module of sections on each QRSP chart. Hence its cosimplicial section complexes have a common lower cohomological bound, and descent identifies their totalization with $\overline{\Delta}_V\{k\}$.

After derived base change by ψ , the same charts remain derived affine and the local line-section complexes become

$$\psi^*(\overline{\Delta}_{R^a}\{k\}).$$

The only issue is whether ψ^* may be interchanged with the infinite totalization. The normalized cosimplicial degree a is a direct summand of the a th term and appears in totalization with shift $[-a]$. Since the unbased terms are uniformly connective, the fibers of the partial-totalization tower become arbitrarily connective. The functor ψ^* has Tor amplitude $[-1, 0]$ by [theorem 8.2](#); therefore the totalization case of [theorem 8.1](#) applies. Flat-local descent on the derived base change then gives (67). All identifications are induced by the twist-graded prismatic multiplication, so they are multiplicative in k .

There is one further derived distinction. Formula (67) computes the pullback of the integral Hodge–Tate line. By definition

$$\psi^*H_k = \psi^*(\overline{\Delta}_V\{k\}/p) \simeq \psi^*(\overline{\Delta}_V\{k\}) \otimes_{\mathbf{Z}_p}^{\mathbf{L}} \mathbf{F}_p.$$

Although the scalar map already lands in the characteristic- p ring A , this derived reduction can have a nonzero Tor term and therefore cannot be omitted. The canonical reduction map is multiplicative, so inverse sections of the integral line remain inverse after passing to $\psi^*H_{\pm 1}$. $\square$

Proposition 8.7 (Frobenius-pulled BK-line unit). *There is a unit*

$$\eta \in H^0(\psi^* H_1)$$

with inverse in $H^0(\psi^ H_{-1})$. Consequently*

$$\psi^* \mathcal{E}_k \simeq \psi^* \mathcal{E}_\ell \quad (k, \ell \in \mathbf{Z}). \quad (68)$$

Proof. Let $f : \operatorname{Spec} A \rightarrow \operatorname{Spf} R$ be given by

$$T \mapsto a^p \bmod p, \quad B \mapsto b^p \bmod p.$$

Consider the base change of the Hodge–Tate Cartier–Witt stack of R along f . The quotient presentation cited above is an equivalence over $\operatorname{Spf} R$; moreover its torsor and the acting divided-power groups are flat over the base. Hence its ordinary termwise base change computes the derived base change used here. For an A -algebra S , the adjoint δ -map $P \rightarrow W(S)$ sends

$$T \mapsto [a^p], \quad B \mapsto [b^p].$$

Since $a^{p^2} = 0$ in A , it sends E to p . Thus the pulled Cartier–Witt divisor is literally the standard divisor $p : W(S) \rightarrow W(S)$.

In the quotient presentation of [3, Proposition 9.5], the torsor of splittings is therefore

$$\{\tau \in W(S) : p\tau = p\} = 1 + W(S)[F] = \mathbf{G}_m^\sharp(S).$$

The divided-power tangent action is controlled by

$$dE = -pT^{p-1}B dT - T^p dB.$$

After applying f , both coefficients vanish in A : the first because $p = 0$, the second because $a^{p^2} = 0$. Hence the tangent group acts trivially on this torsor. If y denotes the zeroth Witt coordinate of τ , then y is invertible and

$$y(u\tau) = u_0 y(\tau)$$

for $u \in \mathbf{G}_m^\sharp$. This is a nowhere-vanishing equivariant section of the character defining the Breuil–Kisin line (or of its dual, according to the sign convention). Thus that line is trivial after the base change f .

Pull this trivialization along $R \rightarrow V$ to

$$Z = \operatorname{Spec} A \times_{\operatorname{Spf} V, \psi} (\operatorname{Spf} V)^{\operatorname{HT}}.$$

By [theorem 8.6](#), its twist-graded global sections are the derived ψ -pullbacks of the integral Hodge–Tate Breuil–Kisin lines, compatibly with multiplication. The geometric trivialization therefore gives inverse sections in

$$H^0 \psi^*(\overline{\Delta}_V \{1\}) \quad \text{and} \quad H^0 \psi^*(\overline{\Delta}_V \{-1\}).$$

Transport them along the canonical multiplicative derived reduction to $\psi^* H_1$ and $\psi^* H_{-1}$. The last step is essential: derived reduction modulo p can retain a Tor term even after the scalar map has landed in A .

Now apply [theorems 8.3](#) and [8.4](#). Multiplication by η identifies every adjacent pair of base-changed filtered twists and hence every adjacent pair of defects. Iteration gives (68). $\square$

9 Vanishing of the Nygaard-completion defects

We can now combine the two independent hard inputs: the endpoint theorem of [theorem 7.5](#) and the twist periodicity of [theorem 8.7](#).

Theorem 9.1 (Defect vanishing). *Let V be a nondiscrete complete rank-one valuation ring of mixed characteristic. Then*

$$\mathcal{E}_k(V) = 0 \quad \text{for every } k \in \mathbf{Z}.$$

Proof. By [theorem 5.6](#), every $\mathcal{E}_k$ lies in $D^{\geq 1}$. We prove inductively that

$$\mathcal{E}_k \in D^{\geq s} \quad \text{for all } k \tag{69}$$

for every $s \geq 1$. Assume (69) for some s .

Fix a weight $j \geq s + 2$. The truncated Bott–defect comparison [theorem 4.4](#) gives

$$H^s C_j \simeq H^{s-1} \mathcal{E}_{j+p-1} = 0.$$

Hence the Bott map

$$H^{s+1} S_j(V) \longrightarrow H^{s+1} S_{j+p-1}(V) \tag{70}$$

is injective. The same argument applies at every later weight in the Bott tower.

Bhatt–Mathew’s Bott localization theorem identifies that telescope with generic étale cohomology:

$$\operatorname{colim}_{m \geq 0} S_{j+m(p-1)}(V) \xrightarrow{\sim} E_j(V), \tag{71}$$

with the usual coefficient identifications [4, Theorem 5.1]. Since the transition maps on H^{s+1} are injective, the comparison

$$H^{s+1} S_j(V) \longrightarrow H^{s+1} E_j(V)$$

is injective. It is also surjective by [theorem 2.6](#), because $s + 1 < j$. Thus it is an isomorphism for every $j \geq s + 2$.

Now take the single weight $j = s + 2$ and put $k = j + p - 1$. The Bott map on H^{s+1} is an isomorphism, because both comparison maps in that degree are isomorphisms and the generic Bott class is invertible. On H^{s+2} , [theorem 7.5](#) makes the comparison in weight $s + 2$ injective; naturality with the invertible generic Bott class therefore makes

$$H^{s+2} S_j(V) \longrightarrow H^{s+2} S_k(V)$$

injective. The long exact sequence of C_j gives

$$H^{s+1} C_j = 0.$$

Applying [theorem 4.4](#) with $q = j = s + 2$ yields

$$H^s \mathcal{E}_{s+p+1} = 0. \tag{72}$$

Thus this one defect belongs to $D^{\geq s+1}$.

Pull back by ψ . Since ψ has Tor amplitude $[-1, 0]$, (72) implies

$$\psi^* \mathcal{E}_{s+p+1} \in D^{\geq s}.$$

By the periodicity (68), the same bound holds for $\psi^*\mathcal{E}_k$ for every k . For any such k , the induction hypothesis and the one-degree Tor spectral sequence give

$$H^{s-1}(\psi^*\mathcal{E}_k) \simeq \mathrm{Tor}_1^V(H^s\mathcal{E}_k, A; \psi) = 0.$$

The module $H^s\mathcal{E}_k$ is annihilated by p . The detection statement in theorem 8.2 therefore forces $H^s\mathcal{E}_k = 0$. Hence every defect lies in $D^{\geq s+1}$, closing the induction.

Thus $\mathcal{E}_k$ is arbitrarily connective. Therefore all of its cohomology groups vanish, so $\mathcal{E}_k = 0$. $\square$

Remark 9.2 (No circularity). *The endpoint theorem does not use Nygaard completeness or defect vanishing: it is proved from Tor-unital excision, the universal degree-three class, and the all-degree Milnor normal form. Conversely, the prismatic weak- F and separation arguments do not use the endpoint. They meet only in the single step (72) of the induction above.*

10 Proof of the all-weight comparison

Proposition 10.1 (Complete rank one). *For every complete rank-one mixed-characteristic valuation ring V and every $n \geq 0$,*

$$\mathrm{cofib}(S_n(V) \rightarrow E_n(V)) \in D^{\geq n}.$$

Proof. Assume first that V is nondiscrete and $n \geq 1$. By theorem 9.1, all completion defects vanish. The truncated Bott–defect comparison therefore gives

$$\tau^{\leq j-1}C_j = 0 \quad (j \geq 1).$$

Hence multiplication by v_1 is injective on $H^qS_j(V)$ whenever $q \leq j$. Apply this along the Bott tower beginning in weight n . The localization equivalence (71) shows that

$$H^qS_n(V) \longrightarrow H^qE_n(V)$$

is injective for every $q \leq n$. For $0 \leq q < n$, it is surjective by theorem 2.6. For $q < 0$, the étale target is zero and injectivity forces the source to vanish. Thus the comparison is an isomorphism below n and injective in degree n .

If V is discrete, it is a complete DVR, hence regular noetherian and F -smooth. The same statement is exactly the regular comparison theorem [4, Theorem 4.15 and Proposition 5.2].

For $n = 0$, both weight-zero theories are connective and the map on H^0 is the identity of $\mathbf{F}_p$ on the connected spectra of V and $V[1/p]$; see the weight-zero identification in [2, Section 8.4]. Thus the cofiber lies in $D^{\geq 0}$. $\square$

Proof of theorem 1.1. Fix p and n . Proposition 10.1 proves the required connectivity for every complete rank-one mixed-characteristic valuation ring. Proposition 2.1, i.e. the v -excision reduction of Bachmann–Elmanto–Morrow, transports this exact statement to every henselian mixed-characteristic valuation ring. No condition remains on the rank, value group, completeness, or on the prime p . $\square$

References

- [1] T. Bachmann, E. Elmanto, and M. Morrow, $\mathbf{A}^1$ -invariant motivic cohomology of schemes, arXiv:2508.09915.
- [2] B. Bhatt and J. Lurie, *Absolute prismatic cohomology*, arXiv:2201.06120.
- [3] B. Bhatt and J. Lurie, *The prismaticization of p -adic formal schemes*, arXiv:2201.06124.
- [4] B. Bhatt and A. Mathew, *Syntomic complexes and p -adic étale Tate twists*, arXiv:2202.04818.
- [5] B. Bhatt, M. Morrow, and P. Scholze, *Integral p -adic Hodge theory*, Publ. Math. Inst. Hautes Etudes Sci. **128** (2018), 219–397; arXiv:1602.03148.
- [6] B. Bhatt, M. Morrow, and P. Scholze, *Topological Hochschild homology and integral p -adic Hodge theory*, Publ. Math. Inst. Hautes Etudes Sci. **129** (2019), 199–310; arXiv:1802.03261.
- [7] B. Bhatt and P. Scholze, *Prisms and prismatic cohomology*, Ann. of Math. (2) **196** (2022), no. 3, 1135–1275; arXiv:1905.08229.
- [8] O. Gabber and L. Ramero, *Almost Ring Theory*, Lecture Notes in Mathematics 1800, Springer, 2003; arXiv:math/0201175.
- [9] T. Geisser and M. Levine, *The K -theory of fields in characteristic p* , Invent. Math. **139** (2000), 459–493.
- [10] A. Hattori, *A foundation of torsion theory for modules over general rings*, Nagoya Math. J. **17** (1960), 147–158.
- [11] M. Kerz, *Milnor K -theory of local rings with finite residue fields*, J. Algebraic Geom. **19** (2010), 173–191.
- [12] M. Lueders and M. Morrow, *Milnor K -theory of p -adic rings*, arXiv:2101.01092.
- [13] Z. Mao, *Frobenius–Witt cotangent complex for derived rings*, arXiv:2608.04487v3.
- [14] M. Morrow, *Pro unitality and pro excision in algebraic K -theory and cyclic homology*, J. Reine Angew. Math. **736** (2018), 95–139; arXiv:1404.4179.
- [15] G. Tamme, *Excision in algebraic K -theory revisited*, Compos. Math. **154** (2018), 1801–1814; arXiv:1703.03331.
- [16] L. Tang, *Syntomic cycle classes and prismatic Poincaré duality*, arXiv:2210.14279.
- [17] W. van der Kallen, *The K_2 of rings with many units*, Ann. Sci. École Norm. Sup. (4) **10** (1977), 473–515.
- [18] V. Voevodsky, *On motivic cohomology with $\mathbf{Z}/\ell$ -coefficients*, Ann. of Math. (2) **174** (2011), no. 1, 401–438; arXiv:0805.4430.